

XYANA2000 Virtual PC

- CNC Machine -

Multipurpose Measurement Program

Instruction Manual (Drills)

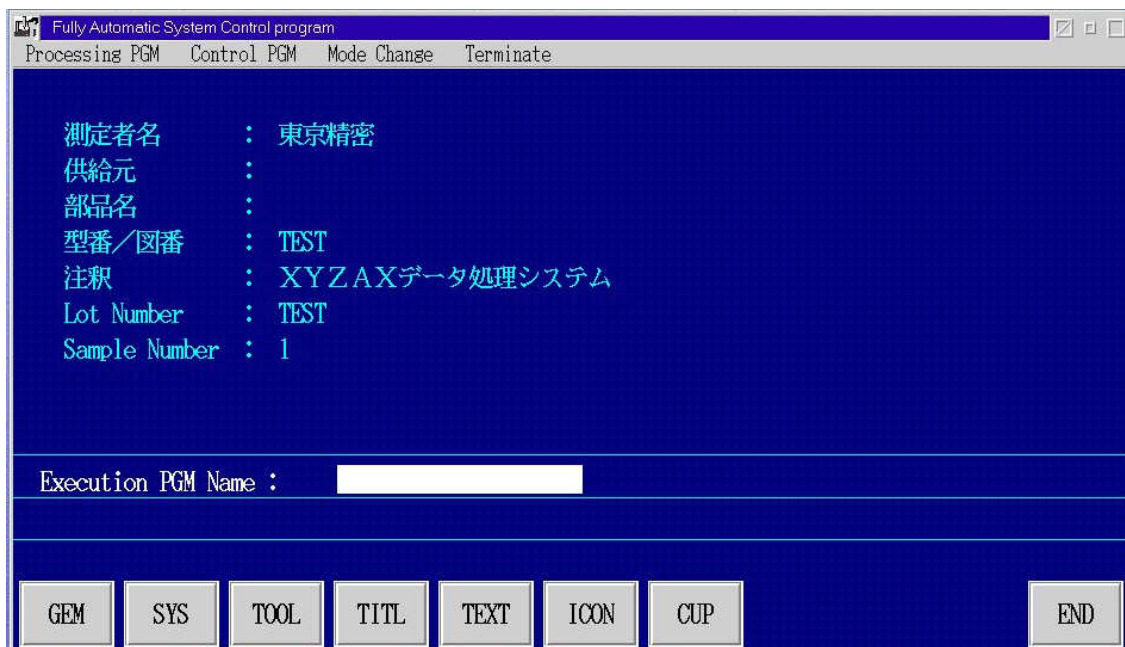
TOKYO SEIMITSU CO., LTD

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1. How to start up system

- 1) Turn on the controller main breaker.
- 2) Turn on the controller switch.
- 3) Check that the X, Y, and Z lamps on the operation board are turning on and turn the controller switch to "DRIVE". Two pilot lamps turn on.
- 4) Turn on the printer power.
- 5) Turn on the monitor power.
- 6) Turn on the computer power.
- 7) Window starts up and the log-in screen appears.
Leave the user name and the password as they are and click .
- 8) The <XYANA2000> icon appears on the desktop. Double-click the icon.
- 9) Virtual Box screen appears and XYANA2000 starts up. Only when PH-9 (or PH10) is connected, PH9 is initialized. A message "Move the PH-9 at a safe place and press ENTER key" appears. Press .
- 10) A message "Home processing is executed after shifting to the home position (upper part of left rear) and press ENTER" appears. Press .
- The probe head automatically shifts to the machine starting point.
- 11) It is displayed as "Execution program name:" and each program is ready to start.



* Execute the following operation as necessary.

(1) To register measurement title information, press TITL on the screen.

Use left/right arrow keys ( ) for the menu and up/down arrow keys ( ) for each item.

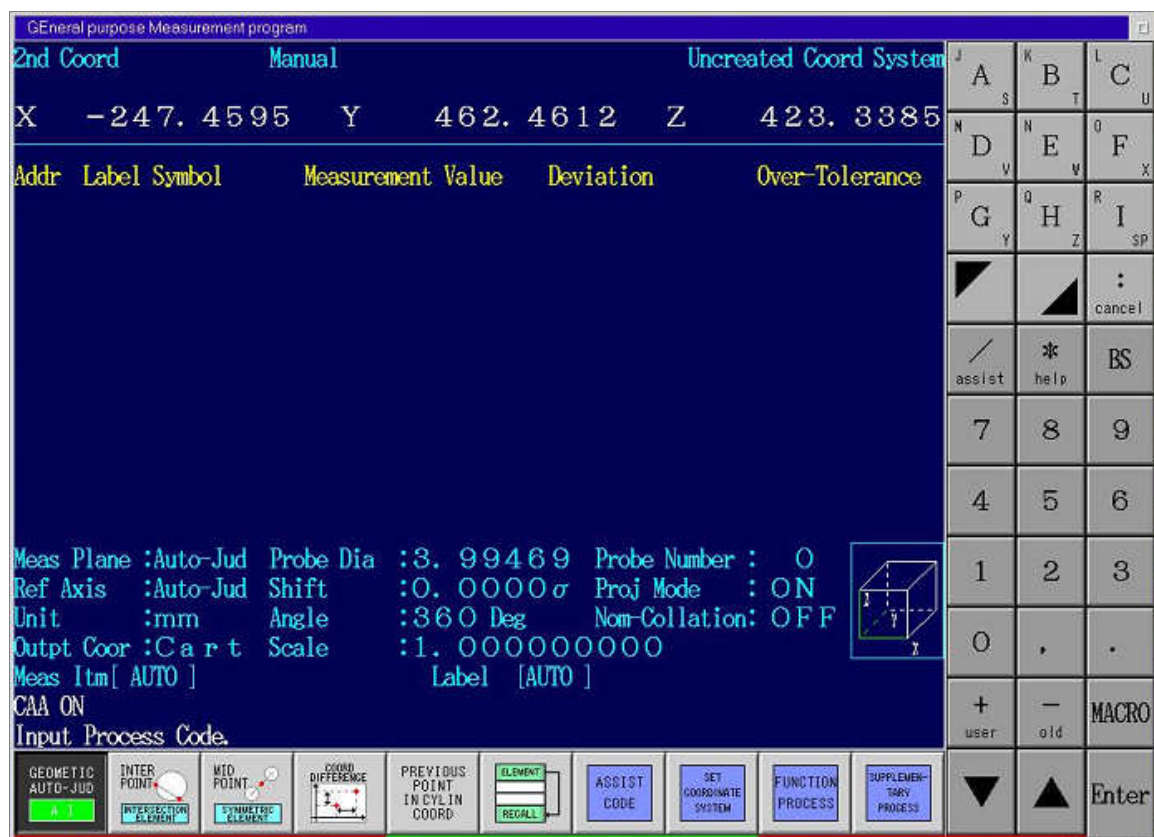
(2) To register the system information, press **SYS** on the screen.

Here, connected device information, measurement information registration, and the CNC controller can be registered.

Use   arrow keys for the menu and   arrow keys for each item.

12) Press **GEM** on the screen to start up the Multi-purpose measurement program.

When the Multi-purpose measurement program starts up, the display shows as the following figure.



<When the DRIVE cannot be entered>

When the drive switch cannot be entered, the following causes may be conceivable. Please check.

- (1) Does the supply air reach the regulated pressure?
- (2) Isn't the emergency stop button pressed?
- (3) Do any of the X, Y, and Z axes reach the end position?

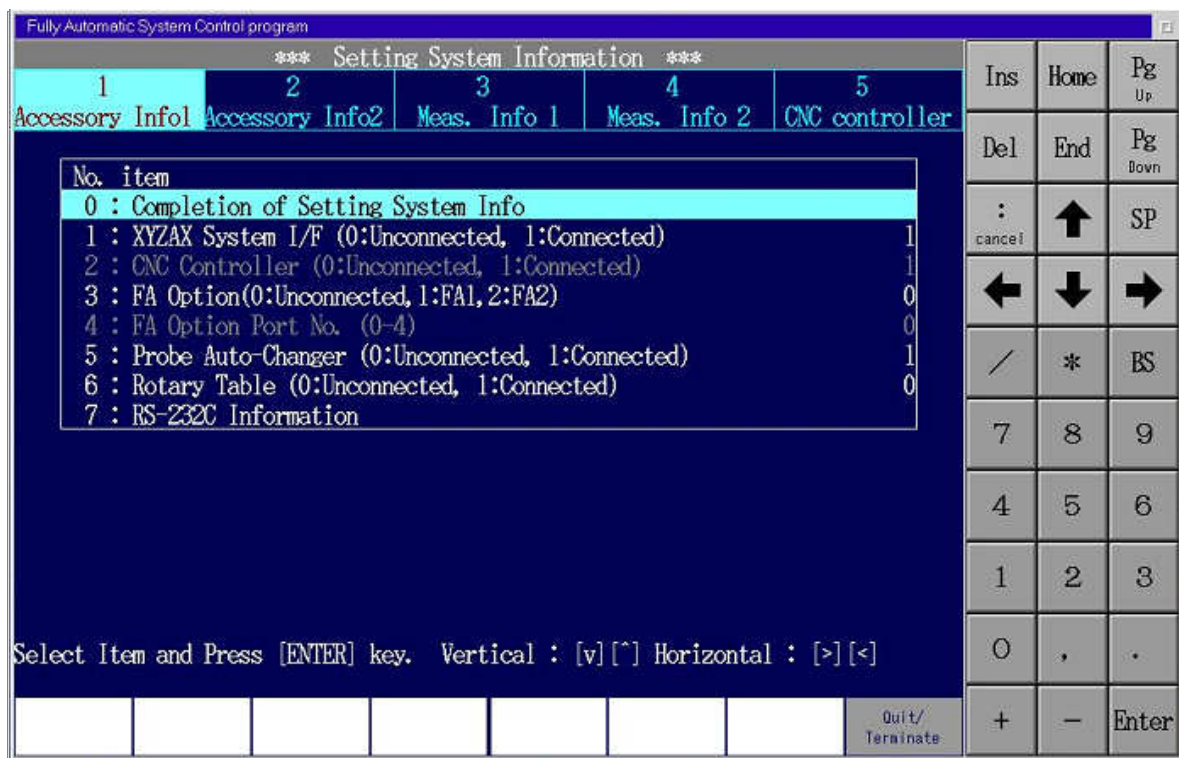
<Notes for startup >

In the following cases, the system connection may not be made correctly. Please check each setting.

- (1) When the Multi-purpose measurement program starts up, the XYZ counter on the upper part of the screen is standing still.
- (2) No points are entered even executing probing.

How to check the setting

- (1) Exit the Multi-purpose measurement program (Select End from the icon menu or input "B99" and press ENTER).
- (2) The initial screen reappears. Select SYS (system information registration).
- (3) The following screen appears. Check that the settings of the article No.1 "XYZAX system I/F (Interface)" and the article No.2 "CNC controller" are set to the connection No.1.
- (4) When it is set to No.0 (unconnected):
 - 1) Place the cursor to the article No. 1 and press ENTER.
 - 2) Input "1" and press ENTER (to connect).
 - 3) Set the article No.2 in the same way.
 - 4) Place the cursor to the article No. 0 (end) and press ENTER.



If the phenomenon still remains even checking the above matters, please make an inquiry.

MEMO

2. How to exit the system

- 1) Exit the Multi-purpose measurement program. Press End on the icon menu. Or, input "B99" and press **ENTER**.

A message "Are you sure to finish the Multi-purpose measurement (0: No 1: Yes) " appears. Input "1" and press **ENTER**.

- 2) The following screen appears. Press **END**.

Fully Automatic System Control program

Processing PGM Control PGM Mode Change Terminate

測定者名 : 東京精密
供給元 :
部品名 :
型番/図番 : TEST
注釈 : XYZAXデータ処理システム
Lot Number : TEST
Sample Number : 1

Execution PGM Name :

GEM SYS TOOL TITL TEXT ICON CUP END

- 3) From the right click menu on the OS/2 desk top screen, select "End of OS/2" and press OK.
- 4) After a few seconds, the following message appears on the screen.

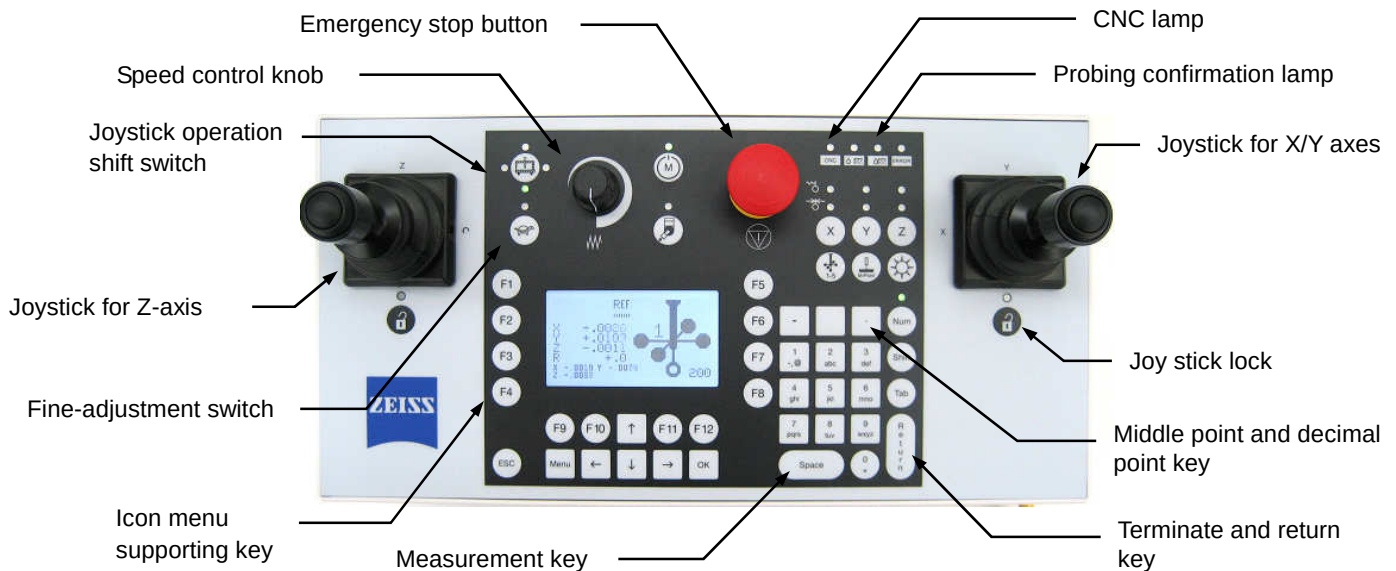
The system exit process is finished.
Turn off the computer power or use
"Ctrl+Alt+Del" keys to restart the system.

* At this moment, never press the computer power-on button.

- 5) Keep pressing the **Ctrl** key at the right of the keyboard and press **Q** (for XP press right ALT+F4 keys).
- 6) Windows desktop screen reappears. From the **Start**, select **Shutdown**.
After a few seconds, the Windows is terminated and the computer power is turned off automatically.
- 7) Turn off the monitor power.
- 8) Turn off the printer power.
- 9) Turn off the controller switch (turning left).
- 10) Turn off the controller main breaker.

MEMO

3. Operation board



◆ Joy stick and CNC measurement speed control knob

Travel speed of the Joystick and CNC measurement speed can be controlled.

◆ Joystick operation shift switch

The viewpoint of the operative position is changed to what is seen from the right/left or rear.

◆ Fine-adjustment switch

The joystick operational speed is changed to 2.5% of the setting value. This is used when using microscope and the like.

◆ Icon menu supporting key

F1 - **F10** keys correspond to the icon menus currently displayed on the monitor.

F11 corresponds to the down arrow key () while F12 corresponds to the up arrow key ().

◆ Emergency stop button

Pressing this button enables the emergency stop and locks the three axes. Turn the button in the arrow direction in order to raise the emergency stop button.

A message "Turn on the motor switch and press ENTER" appears on the monitor. Turn the controller switch into "DRIVE" and press ENTER key.

◆ Measurement key

The key is used when using a fixed probe, a microscope, and inputting measurement points.

◆ Terminate and return key

The key is used as a terminate key at the end of multi-points measurement.

In addition, the key is used as the ENTER key for inputting characters.

◆ Middle point and decimal point key

The key is used for inputting middle points for teaching measurement.

In addition, the key is used as the decimal point key for inputting characters.

◆ **Joystick lock (below the right and left joystick)**

This is a button used for locking (turning on the light)/unlocking (turning off the light) the joystick.

〈 **Display Key Operation** 〉

J A S	K B T	L C U
M D V	N E W	O F X
P G Y	Q H Z	R I S P
		② : cancel
③ / assist	④ * help	B S
7	8	9
4	5	6
1	2	3
0	,	.
⑤ + user	⑥ — old	⑦ MACRO
		ENTER

① Alphabetic key

The key is used for inputting processing code number, and program file names. The alphabet flips by pressing  and  keys.

② Cancellation key

The key is used for cancelling a process.

③ Assist key

When executing coordinate system assist, press this key to select an assist No.

④ Help key

Pressing this key shows a list of the process codes on the monitor. To cancel, press **ENTER** key.

⑤ User code execution key

The key is used for executing a user code. Press this key and input the code number to execute.

⑥ Old code key

By pressing this key, input of old code is awaited.

⑦ Macro key

The key is used for executing a part program registered in the macro registration.

⑧ Up-down shift key

The key is used for shifting the icon menus.

⑧

4. How to input process codes

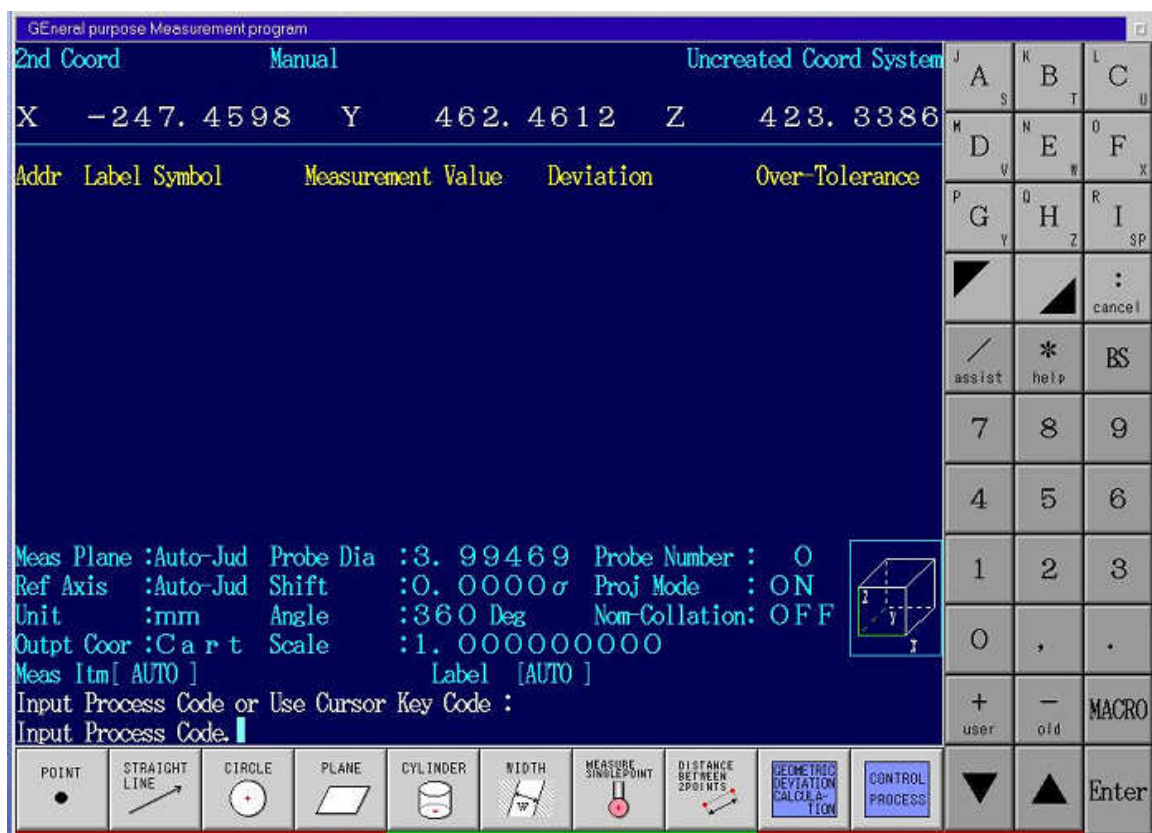
At the startup, as "Geometry automatic recognition (AI function)" is turned on, it is possible to proceed a measurement without selecting the menu.

In addition, when starting a measurement with the "Geometry automatic detection (AI function)" turned off, it is absolutely necessary to input a corresponding process code.

*Refer to CS10014-R107 Instruction Manual for the "Geometry automatic detection (AI function)".

There are two methods for inputting process codes as follows.

(1) Select the icon menu at the lower part of the screen. Directly touch the monitor to select the icon.



The icon menu also can be selected by using the **F1** to **F12** keys on the operation panel. Refer to P.7.

- *1. The icon menu can be shifted by using the  and  (Down/Up arrow) keys.
- *2. When selecting a double framed icon menu in the basic menu (coordinate system setting, function process, etc.), the "Child menu" for respective process is displayed.
If the "Return menu" in the child menu is pressed, the Basic menu appears again.

(2) Selecting a process code by inputting numbers.

The code is input from a full-keyboard and a monitor.

For each process code number, refer to the icon menu table.

By pressing the on the monitor, it is possible to check the process code on the monitor first, and input the process code.

The process code is grouped by menu; A**, B**.C**.F**, and K**. Two digit number are input in the "★★".

〈 Example 〉 When specifying a circle (process code: A02).

*** For cancelling a process**

In the Multi-purpose measurement program, if an operator is required to use the key for input, the message is displayed in query form.

At this time, it is possible to cancel the ongoing process by inputting ":(colon)" and pressing .

5. Coordinate system and flat surface

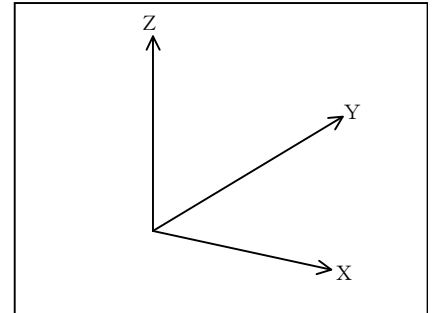
1) Coordinate system

The one composed of X, Y, and Z axes is called coordinate system. For the measurement machine, the axes are fundamentally set as follows: X axis; right/left direction, Y axis; front/rear direction, and Z axis; height.

Five coordinate systems are used for data processing.

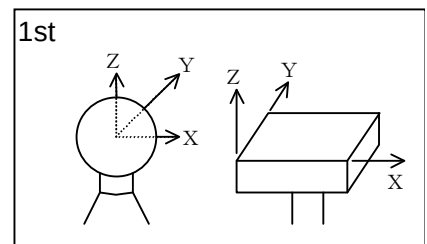
0th coordinate system

Being parallel to the machine coordinate system, the probe center point at the time of turning on the system interface power, becomes the starting point.



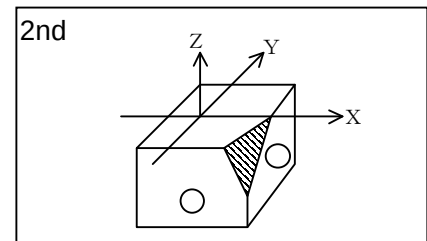
1st coordinate system

Being parallel to the machine coordinate system, the position of the ball starting point block or the square starting point block becomes the starting point.



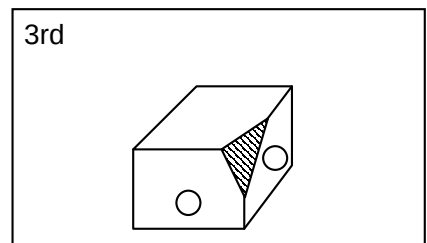
2nd coordinate system

It is the coordinate system of X, Y, and Z axes which is set according to the standard of the actual object to be measured, and the data are derived from this coordinate system.



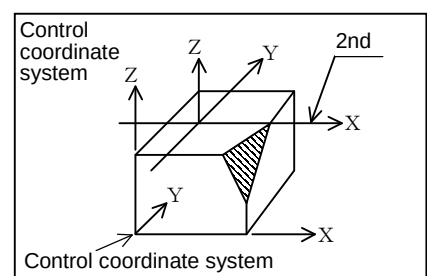
3rd coordinate system

It is a projection surface necessary for calculating coordinate value obtained by probing, by projecting the data on other than the 2nd coordinate system measurement flat surface.



4th coordinate system(control coordinate system: CNC)

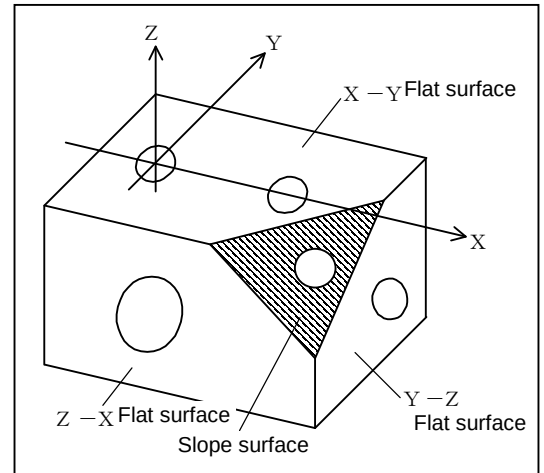
It is a coordinate system for probe to move under the command of the controller. That is, it is a reference coordinate system when creating a part program in the teach mode and executing the part program (auto mode). The middle point information in the part program and the probing point information are expressed in the coordinate values (X, Y, and Z) from this coordinate system.



2) Flat surface

For measuring a work, in general, the reference (2nd coordinate system) suited for the work is set. Depending on the place for measurement, one of the four surfaces including X-Y flat surface, Y-Z flat surface, Z-X flat surface, and slope surface is selected.

For non slope surfaces, it is not necessary to select a flat surface to be measured thanks to the measurement surface automatic detection function so that the measurement can be proceeded easily (It is set to the automatic detection at the initial start).



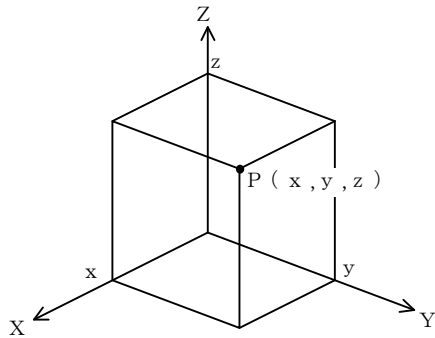
Relation between measurement flat surface and axis

Measurement flat surface	1st axis	2nd axis	3rd axis
X-Y	X-axis	Y-axis	Z-axis
Y-Z	Y-axis	Z-axis	X-axis
Z-X	Z-axis	X-axis	Y-axis
Y-X	Y-axis	X-axis	Z-axis
Z-Y	Z-axis	Y-axis	X-axis
X-Z	X-axis	Z-axis	Y-axis

6. How to represent measurement data

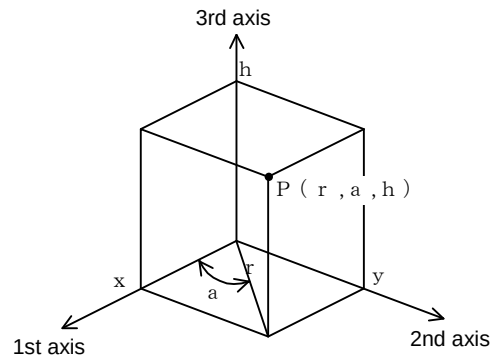
The following three types are used for representing the measurement data.

① Using rectangular coordinate



The measurement point is output by X, Y, and Z.

② Using cylindrical coordinate (flat surface polar coordinate)



$P(r, a, h)$

$r : \sqrt{x^2 + y^2}$ (Flat surface distance)

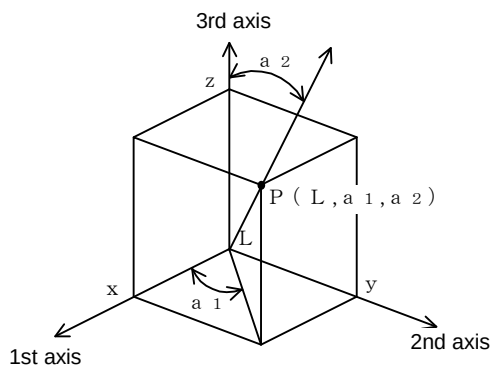
a : The angle formed with the 1st axis

h : 3rd axis value

The measurement point is output by radius vector R, AXY (angle), and Z.

↓
It changes by the flat surface

③ Using polar coordinate (solid polar coordinate)



$P(L, a1, a2)$

$L : \sqrt{X^2 + Y^2 + Z^2}$ (Solid distance)

$a1$: The angle formed with the 1st axis

$a2$: The slant angle from the 3rd axis

The measurement point is output by radius vector L, AXY (angle) and AZ.

*A rectangular coordinate is provided at the startup.

To shift into the cylindrical coordinate or the polar coordinate, use the output coordinate change (B10) of the auxiliary processing.

To output the measurement point immediately before in cylinder and polar coordinate, use the previous point cylinder output (B38) and the previous point polar output (B39) of the auxiliary process.

MEMO

7. General measurement procedure

7-1 Measurement using electron probe

- 1) How to start up the system (refer to P.1).
- 2) Fix the ball starting point block and the object to be measured onto the table.

- 3) Restarting measurement



- 4) Calibration



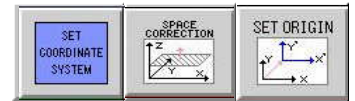
Probe number: Input "0", and press **ENTER**.

Measure at least four points in the ball starting point block, and press Terminate (The 1st coordinate system is set).

- 5) Reference surface setting



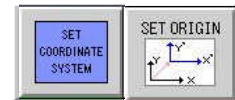
Measure at least three points on the Surface A, and press Terminate.



- 6) Origin point setting



Measure at least three points on the Hole B, and press Terminate.



- 7) Reference axis setting

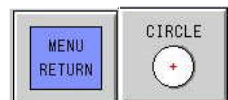


Measure at least three points on the Hole C, and press Terminate.



By the above operations from 5) to 7), the 2nd coordinate system is set.

- 8) Measurement



Measure at least three points on the Hole D, and press Terminate.

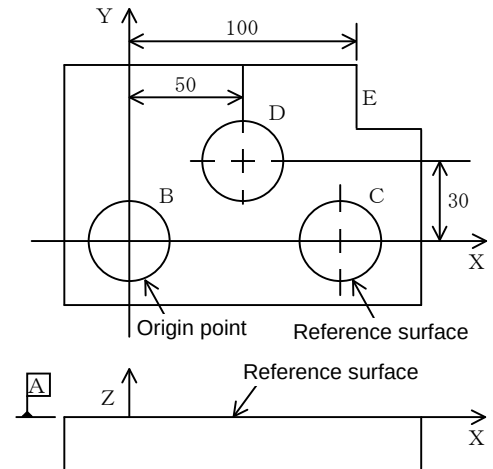
[X = 50 Y = 30]



Measure at least one point on the Edge surface E, and press Terminate.

[X = 100]

- 9) Exiting the system (refer to P.5).

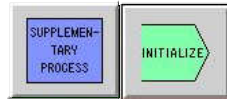


7-2 Measurement using centering microscope and ITV camera system

1) How to start up the system (refer to P.1).

2) Horizontally, set the object to be measured on the table.

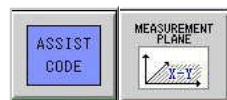
3) Restarting measurement



4) Turning off the geometry automatic detection (AI function)



5) Fixation of measurement flat surface



1 (X Y)

6) Changing probe diameter

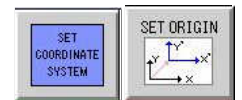


Probe diameter: Input "0"

7) Origin point setting



Set the crossline on the circumference of the Hole A, and press Measurement key.

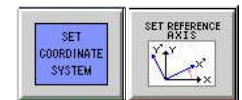


Measure at least three points and press Terminate.

8) Reference axis setting



Set the crossline on the circumference of the Hole B, and press Measurement key.



Measure at least three points and press Terminate.

By the above operations 7) and 8), the 2nd coordinate system is set.

9) Measurement



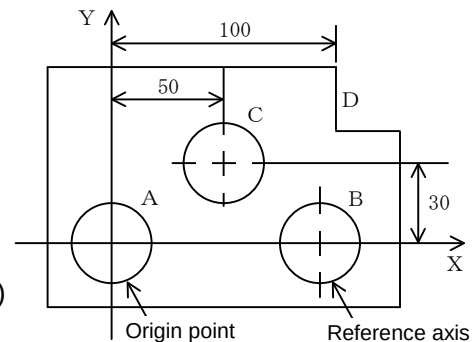
Set the crossline on the circumference of the Hole C, and press Measurement key.

Measure at least three points and press Terminate. [X = 50 Y = 30]



Set the crossline on the Edge surface D, and press Measurement key. [X = 100]

10) Exiting the system (refer to P.5).

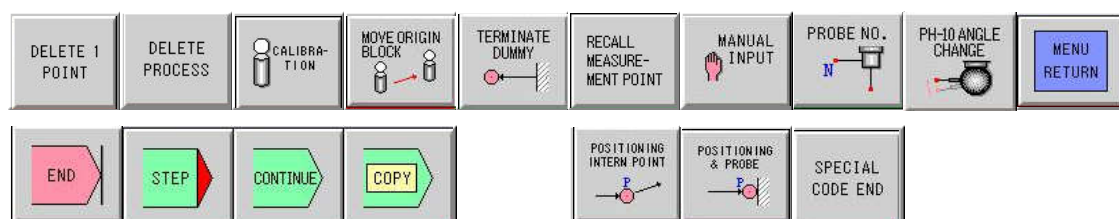


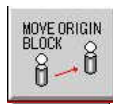
8. Function process

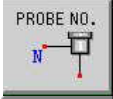
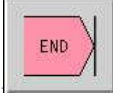
The function process is a process to be executed as an interrupt while a measurement process or an automatic measurement process is underway.

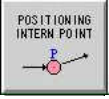
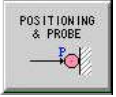
The following shows a detail of such a process.

Child menu of function process : F**



Process code	Content
 (F0)	In the manual mode, one of the probing points immediately before is deleted (It is used when reading a wrong probing point during the measurement). *The content of this process differs depending on the measurement mode. For the measurement mode other than the manual mode, refer to P3 to P8 in the Function chapter (VOL.4).
 (F1)	In the manual mode, all the probing points read are deleted (It is used when noticing that wrong process code is selected while the measurement is underway). *The content of this process differs depending on the measurement mode. For the measurement mode other than the manual mode, refer to P9 to P15 in the Function chapter (VOL.4).
 (F20)	The probe to be used for setting the first coordinate system and the measurement is registered.
 (F21)	It is a process for shifting the Origin point block (It is used when changing the probe direction during measurement or when the calibration is disabled because of interference of the object to be measured).
 (F42)	The multi-point measurement is terminated. The measurement recall or the point input by the manual input become the terminate point (dummy point). (Generally, the "terminate" that terminates the multi-point measurement after automatically calibrating the probe radius share is used. This "dummy terminate" is used for terminating the multi-point measurement that does not calibrate the probe radius share).

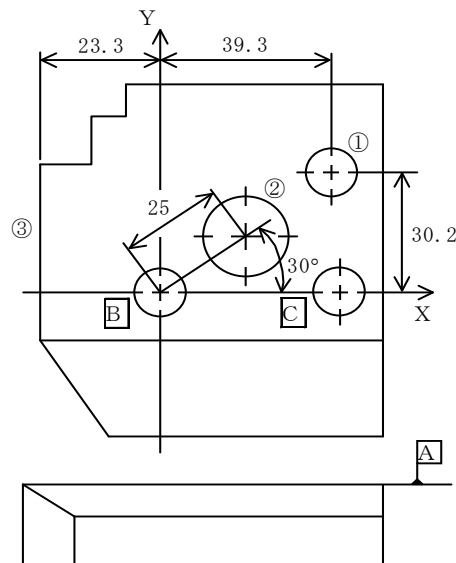
Process code	Content
 (F23)	The already measured point (measurement point) is called.
 (F24)	The coordinate value input from the keyboard is set to the measurement point.
 (F22)	The probe number is changed (It is used for the measurement using two or more probes).
 (F25)	It is changed to the angle by which the PH-9 probe direction is specified. *This process can be executed only when connecting the PH-9.
 (F50)	Execution of user code, execution and registration of repeat, and execution of old codes are exited.
 (F10)	Forcibly shifting to the Auto Step mode or the Edit step mode when executing the Auto mode or the Edit mode. *This process can be executed only in the Auto mode or the Edit mode.
 (F11)	The part program is executed step-by-step basis while the Auto Step mode or the Edit Step mode is underway. *This process can be executed only in the Auto step mode or the Edit step mode.
 (F12)	Shifting into the Auto Step mode or the Edit Step mode again while the Auto Step mode or the Edit Step mode is underway. *This process can be executed only in the Auto step mode or the Edit step mode.
 (F13)	The process copies a part of the part program (specified portion) without executing the program while the Edit Step mode is underway. *This process can be executed only in the Edit step mode.

Process code	Content
 (F40)	The probe is shifted to the coordinate value input.
 (F41)	The probe (measurement) is executed for the coordinate value input.

MEMO

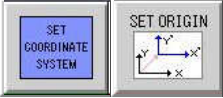
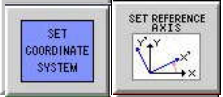
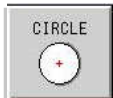
Measurement procedure

9. Exercise 1



Reference flat surface: Surface A
Origin point specification: Hole B
Reference axis specification; Hole C

Address	Icon and operation method	Remarks	Nominal value
	The second and the third coordinate system are cleared, the coordinate system address is set to "C001", and element address is set to "A0001". The printer begins the new page and prints out the title.		
	Probe number: 0 ENTER At least four points in the ball Origin point block is measured and press Terminate. Probe number: input "0", probe diameter: input "_", standard deviation: input "_". Is it accepted? (0: No, 1:Yes): Input "1" and press ENTER .	Setting of 1st coordinate system. The probe diameter is automatically registered.	
A0001	Measure at least three points on the Surface A, and press Terminate.		
C0001 C0002		Setting the surface A as the reference surface (XY surface). The perpendicular line of Surface A is parallel to the Z axis. On the Surface A becomes Z=0.	

Address	Icon and operation method	Remarks	Nominal value
A0002	 Measure at least three points on the Hole B, and press Terminate.		Diameter =12
C0003		Hole B center is assumed to be the Origin point.	
A0003	 Measure at least three points on the Hole C, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	Diameter =12
C0004		The coordinate system is rotated so that the X axis passes through the Hole C center.	
A0004	 Measure at least three points on the Hole ①, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	Diameter =12 X=39.3 Y=30.2
A0005	 Measure at least three points on the Hole ②, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	Diameter =28
A0005		Outputting cylinder from the Hole ② center	Radius vector R =25 AXY=30°
A0006	 Measure at least one point on the Edge surface ③, and press Terminate.		X= -23.3

A0001	A05	PLN	Z	-8.352				
			AVXY	-44.05.17	AVZ	0.04.46		

C0001 A80 COD2 Reference surface set (X Y)

C0002 A81 COD2 Origin point set

A0002	A02	CIR	Diameter	11.979	Radius	5.989		
			X	-151.588	Y	-416.551		

C0003 A81 COD2 Origin point set

A0003	A02	CIR	Diameter	11.989	Radius	5.995		
			X	43.249	Y	-2.431		

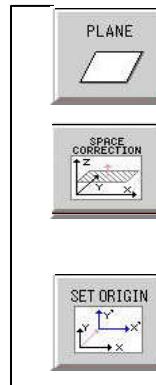
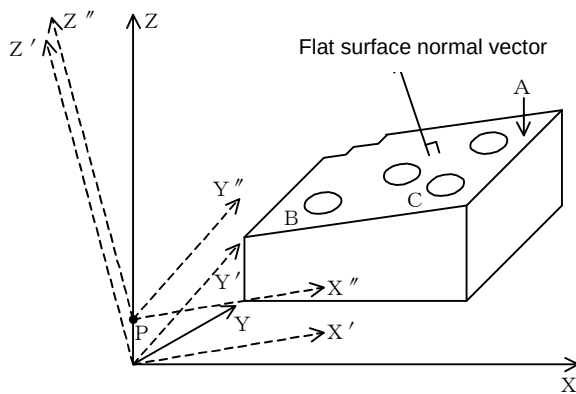
C0004 A82 COD2 Reference axis set(X)

A0004	A02	CIR	Diameter	11.997	Radius	5.998		
			X	39.331	Y	30.176		

A0005	A02	CIR	Diameter	27.882	Radius	13.941		
			X	21.678	Y	12.419		

A0005	B38	CIR	Radius vector R	24.984	AXY	29.48.29	Z	0.000
A0006	A00	PNT	X	-23.272				

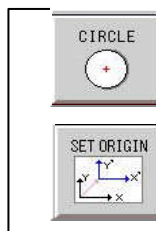
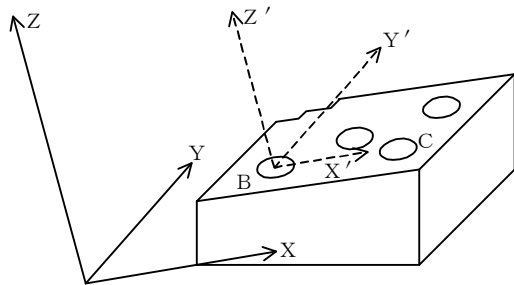
Explanation of exercise 1



The intersection P between the flat surface and the Z axis, and the normal vector for the flat surface are obtained.

Flat surface normal vector and the Z' axis become parallel each other (X', Y', Z').

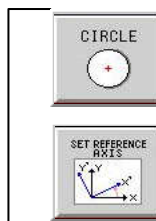
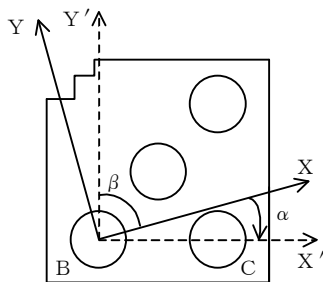
By the automatic detection, the measurement flat surface is selected as a flat surface whose normal vector closest to vertical against the coordinate flat surface of the 1st coordinate system (X, Y, Z), so that the surface A becomes XY surface. The Origin point shifts in parallel to the P point (X'', Y'', Z'').



Measuring the Hole B.

The circle center is obtained.

The Origin point shifts in parallel to the circle center (X', Y', Z').



Measuring the Hole C.

The circle center is obtained.

The reference axis (X axis) is rotated so as to pass through the Hole C center (X', Y', Z').

By the reference axis automatic detection, the axis of smaller angle when rotating the coordinate system is assumed to be the reference axis. Because α is smaller than β , the reference axis that passes through the Hole C center becomes the X axis.

Previous point cylinder output (B38)

The measurement point immediately before is re-output in the cylindrical coordinate value.

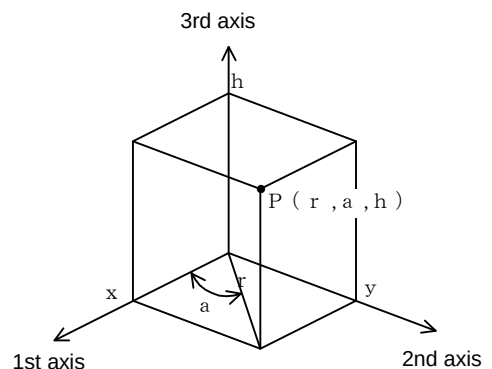
P (r, a, h)

r : $\sqrt{x^2 + y^2}$ (Radius vector R)

a : The angle formed with the 1st axis (AXY)

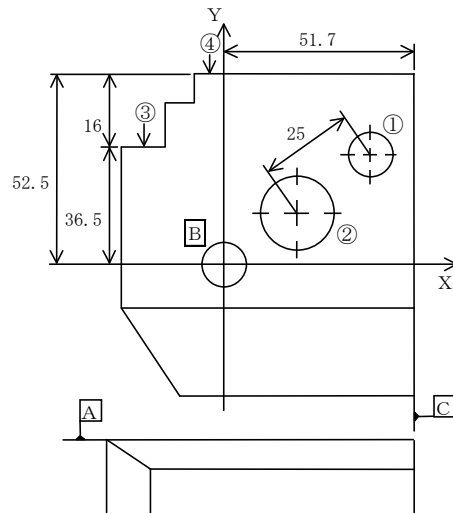
h : The third axis value

It changes by the flat surface



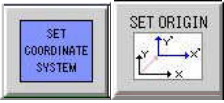
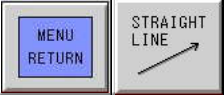
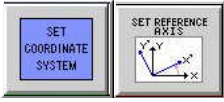
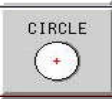
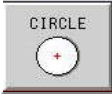
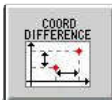
Measurement procedure

10. Exercise 2



Reference flat surface: Surface A
Origin point specification: Hole B
Reference axis specification; Edge C

Address	Icon and operation method	Remarks	Nominal value
	 		
	   Probe number: 0 ENTER At least four points in the ball Origin point block is measured and press Terminate. Probe number: input "0", probe diameter: input "_", standard deviation: input "_". Is it accepted? (0: No, 1:Yes): Input "1" and press ENTER .	Setting of 1st coordinate system. The probe diameter is automatically registered.	
A0001	 Measure at least three points on the Surface A, and press Terminate.		
C0001 C0002	  	Setting the surface A as the reference surface (XY surface). The perpendicular line of Surface A is parallel to the Z axis. On the Surface A becomes Z=0.	
A0002	  Measure at least three points on the Hole B, and press Terminate.		Diameter = 12

Address	Icon and operation method	Remarks	Nominal value
C0003		Hole B center is assumed to be the Origin point.	
A0003	 Measure at least two points on the Edge C, and press Terminate.		
C0004		Y axis is set in parallel to the Edge C.	
A0004	 Element address number: A3 ENTER (Or - 1)	The Edge C is recalled	X=51.7
A0005	 Measure at least three points on the Hole ①, and press Terminate.		Diameter =12
A0006	 Measure at least three points on the Hole ②, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	Diameter =28
	 Element address number [-2,-1] : ENTER	The direct distance between the center of the Hole ① and the center of the Hole ② is obtained.	Distance =25
A0007	 Measure at least one point on the Edge surface ③, and press Terminate.		Y=36.5
A0008	 Measure at least one point on the Edge surface ④, and press Terminate.	The icon specification is not required because the point measurement is executed immediately before.	Y=52.5
	 Element address number [-2,-1] : ENTER	The coordinate difference between the Edge surface ③ and the Edge surface ④ is obtained.	Difference Y=16

A0001 A05 PLN Z -9.566
 AVXY 177.40.08 AVZ 0.03.00

C0001 A80 COD2 Reference surface set(XY)
 C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 12.051 Radius 6.025
 X -277.023 Y -211.151

C0003 A81 COD2 Origin point set

A0003 A01 LIN X 51.641
 AVXY 89.50.19

C0004 A82 COD2 Reference axis set(Y)

A0004 A60 LIN X 51.641 Z 0.000
 AVZX -91.23.14 AVY 0.00.00

A0005 A02 CIR Diameter 12.045 Radius 6.022
 X 39.327 Y 30.169

A0006 A02 CIR Diameter 28.079 Radius 14.040
 X 21.647 Y 12.489

A64 Distance 25.003

A0007 A00 PNT Y 36.464

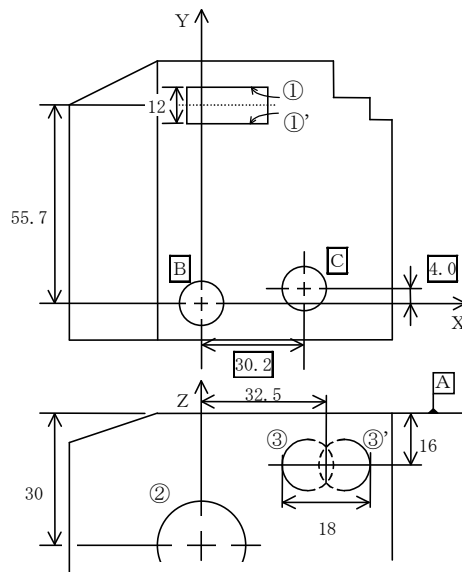
A0008 A00 PNT Y 52.496

A65 Difference X 0.000 Difference Y 16.032 Difference Z 0.000

MEMO

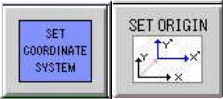
Measurement procedure

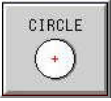
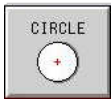
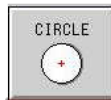
11. Exercise 3



Reference flat surface: Surface A
Origin point specification: Hole B
Reference axis specification; Hole C

Address	Icon and operation method	Remarks	Nominal value
	Probe number: 0,1 ENTER With setting the downward to "0" and backward to "1", measure at least four points in the ball Origin point block for each of them and press Terminate. Probe number: 0 ENTER	Setting of 1st coordinate system. The probe diameter is automatically registered. *It is a method for making a measurement using two or more probes. The probe No.0 is called.	
A0001	Measure at least three points on the Surface A, and press Terminate.		
C0001 C0002		Setting the surface A as the reference surface (XY surface). The perpendicular line of Surface A is parallel to the Z axis. On the Surface A becomes Z=0.	

Address	Icon and operation method	Remarks	Nominal value
A0002	 Measure at least three points on the Hole B, and press Terminate.		Diameter =12
C0003		Hole B center is assumed to be the Origin point.	
A0003	 Measure at least three points on the Hole C, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	
C0004	 Type: Input "4" and press ENTER . 1st axis offset: Input "30.2" and press ENTER . 2nd axis offset: Input "4" and press ENTER .	The reference axis (X axis) is set, so as to let the Hole C center be suitable for the offset amount input.	
A0004	 Element address number: A3 ENTER (Or -1)	It is recalled for confirmation.	X=30.2 Y=4
A0005	 Measure ① at least one point, and press Terminate.		
A0006	 Measure ①' at least one point, and press Terminate.	The icon specification is not required because the point measurement is executed immediately before.	
A0007	 Element address number [-2,-1] : ENTER	The distance: "width", and Y: "width center" are obtained.	Distance =12 Y=55.7
※	 Probe number:1 ENTER	In order to measure the transverse hole, the probe direction is changed into "1".	

Address	Icon and operation method	Remarks	Nominal value
A0008	 Measure at least three points on the Hole ②, and press Terminate.		Diameter = 28 X = 0 Z = -30
A0009	 Measure at least three points on the Hole ③, and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	
A0010	 Measure at least three points on the Hole ③', and press Terminate.	In a similar case, the icon specification is not required because the cycle measurement is executed immediately before.	
	 Element address number [-2,-1] : ENTER	The longest distance between the Hole ③ and the ③' is obtained.	Longest distance = 18
A0011	 Element address number [-2,-1] : ENTER	The middle point of the Hole ③ and the Hole ③' is obtained.	Distance = 6 X = 32.5 Z = -16

Continuous calibration

- Probes are combined depending on the measurement locations. The downward is assumed to be "0" and the other to be "1."

-  Probe number (*.* ,..., *.*) [0] : 0, 1

Example: Inputting "0.2" and pressing → Three units from 0 to 2 are continuously calibrated.
Inputting "0,2" and pressing → Two units of 0 and 2 are continuously calibrated
The number possible to specify: from 0 to 99.

- Measurement item: Calibration (sphere)

Execute the calibration using the No. (0) probe.

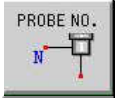
With the probe of the number displayed, measure at least four points in the ball Origin point and press Terminate.

Probe number: "0", probe diameter: "4.9956", and standard deviation: "0.0002".

Is it accepted? (0: No, 1:Yes) [1]:Input "1" and press .

- If repeating the operation (3) for the number specified, the continuous calibration is terminated.

5) The registered probes can be used appropriately for purpose by changing the probe number during the measurement.

Example:  Probe number: 0

```
A0001 A05 PLN   Z           -8.174
              AVXY  -117.40.52   AVZ           0.04.09
```

```
C0001 A80 COD2 Reference surface set(XY)
C0002 A81 COD2 Origin point set
```

```
A0002 A02 CIR   Diameter      11.990      Radius      5.995
              X           -266.669      Y           -304.288
```

```
C0003 A81 COD2 Origin point set
```

```
A0003 A02 CIR   Diameter      11.983      Radius      5.991
              X           30.203      Y           3.709
```

```
C0004 A92 COD2 Reference axis set
```

```
A0004 A60 CIR   Diameter      11.983      Radius      5.995
              X           30.170      Y           3.970      Z      0.000
```

```
A0005 A00 PNT   Y           61.641
```

```
A0006 A00 PNT   Y           49.684
```

```
A0007 A62 PNT   Distance      11.957
              X           0.000      Y           55.663      Z      0.000
```

```
A0008 A02 CIR   Diameter      28.096      Radius      14.048
              X           0.447      Z           -30.050
```

```
A0009 A02 CIR   Diameter      11.958      Radius      5.979
              X           29.426      Z           -16.041
```

```
A0010 A02 CIR   Diameter      11.991      Radius      5.996
              X           35.430      Z           -16.044
```

```
      A63   Shortest distance 5.971      Longest distance 17.979
```

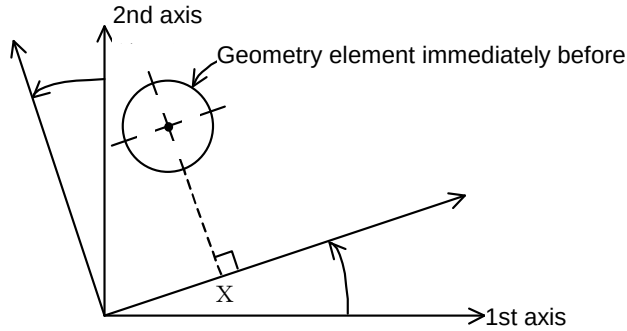
```
A0011 A62 PNT   Distance      6.004
              X           32.429      Y           0.000      Z      -16.042
```

Explanation of exercise 3

〈 Reference axis specification with offset(A92) 〉

Type 1: 1st axis direction offset specification

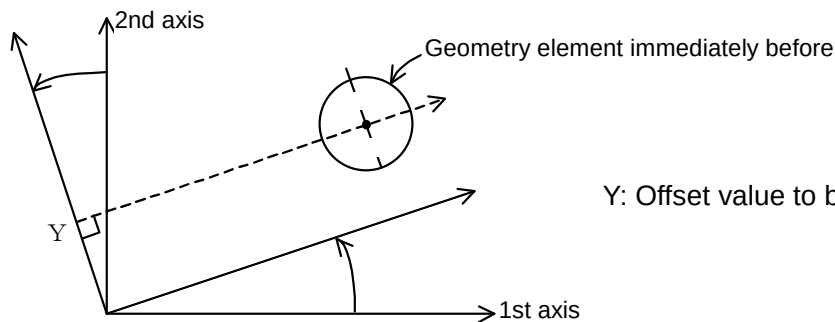
The coordinate system is rotated so that the value of the 1st axis value, among the coordinate value of the measurement point possessed by the geometry element immediately before, becomes the offset value input. The 2nd axis value after the rotation always becomes a positive value.



X: Offset value to be input (1st axis value)

Type 2: 2nd axis direction offset specification

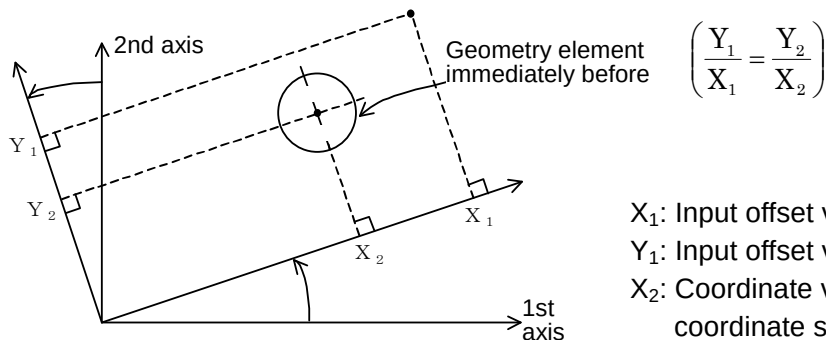
The coordinate system is rotated so that the value of the 2nd axis value, among the coordinate value of the measurement point possessed by the geometry element immediately before, becomes the offset value input. The 1st axis value after the rotation always becomes a positive value.



Y: Offset value to be input (2nd axis value)

Type 3: Two axes direction offset specification (equal ratio)

The coordinate system is rotated so that the ratio of the 1st axis value and the 2nd axis value, among the coordinate value of the measurement points possessed by the geometry element just before, may become equal to the input off set value ratio of the 1st axis value and the 2nd axis value.



X₁: Input offset value (1st axis value)

Y₁: Input offset value (2nd axis value)

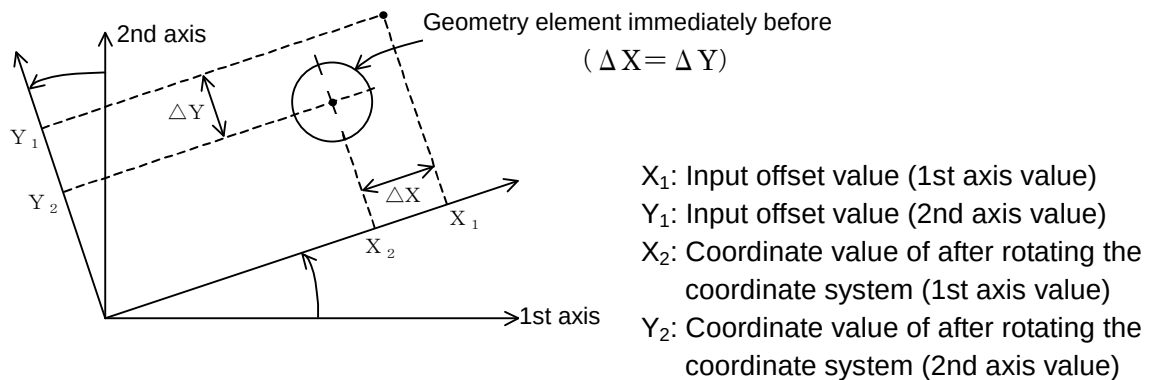
X₂: Coordinate value of after rotating the coordinate system (1st axis value)

Y₂: Coordinate value of after rotating the coordinate system (2nd axis value)

* It is used when the size ratio and the tolerance ratio between the X₁ and the Y₁ are equalized.

Type 4: Two axes direction offset specification (equal error)

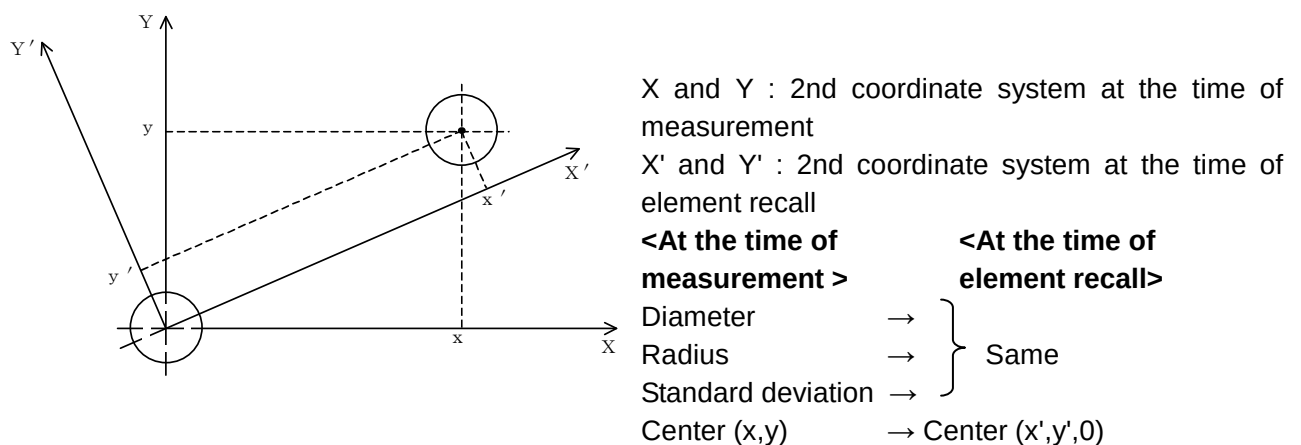
The coordinate system is rotated so that the error on the coordinate system between the coordinate value of the measurement point possessed by the geometry element immediately before and the offset value becomes equal.



* It is used when the tolerance is equalized regardless of the X_1 and the Y_1 sizes.

< Element recall (A60) >

It is a process for re-outputting the already calculated geometry element based on the coordinate system value set at the time of recall.



There are an absolute specification and a relative specification when inputting an element address number.

Absolute specification: The address number of the data you want to recall is directly entered.
 If you want to recall the Hole C data as the exercise 3, input "3".

Relative specification: If data you want to recall is the one immediately before, input "1", and if it is 2nd before, input "-2." Starting from the data immediately before, decrement the number like -1, -2, -3 (Only count the address attached with "A").
 Immediately after a measurement, if you want to recall the Hole C data as the exercise 3, input "-1" as the data is assumed to be the data immediately before.

Short and longest distance (A63)

The short and the longest distance between the geometry elements stored in two address numbers immediately before are obtained.

Because only the distance is obtained, the calculation result is not stored.

The combination of the shortest and the longest distance calculation is shown as follows.

	Point	Straight line	Circle	Oval	Sphere	Flat surface	Cylinder	Cone
Point	D	D	SD LD	×	SD LD	D	SD LD	×
Straight line	—	D	SD LD	×	SD LD	D	SD LD	×
Circle	—	—	SD LD	×	×	×	×	×
Oval	—	—	—	×	×	×	×	×
Sphere	—	—	—	—	SD LD	SD LD	SD LD	×
Flat surface	—	—	—	—	—	D	SD LD	×
Cylinder	—	—	—	—	—	—	SD LD	×
Cone	—	—	—	—	—	—	—	×

×: Inappropriate combination

SD: Shortest distance

—: Same as the opposite combination

LD: Longest distance

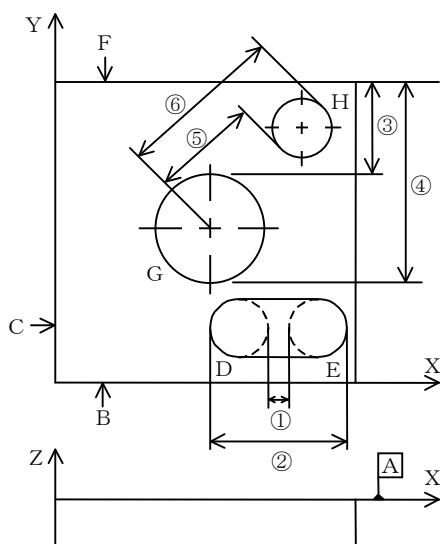
D: Distance

Example of the short and longest distances

1) Circle and circle

2) Straight line and circle

3) Point and circle



		A			
		B			
		C			
1)		D		E	SD=① LD=②
2)		F		G	SD=③ LD=④
3)			—1[ENTER]		H SD=⑤ LD=⑥

(Recalling the Hole G)

Symmetry element (A62)

The symmetry elements (middle point, symmetry line, and symmetric surface) of the geometry elements stored in two address numbers immediately before are obtained.

The result of the output is stored as a geometry element.

The combination of the symmetry element is shown as follows.

	Point	Straight line	Circle	Oval	Sphere	Flat surface	Cylinder	Cone
Point	Point	×	Point **	Point **	Point **	×	×	×
Straight line	—	Straight line	×	×	×	Straight line *	Straight line *	Straight line *
Circle	—	—	Point **	Point **	Point **	×	×	×
Oval	—	—	—	Point **	Point **	×	×	×
Sphere	—	—	—	—	Point **	×	×	×
Flat surface	—	—	—	—	—	Flat surface	Straight line *	Straight line *
Cylinder	—	—	—	—	—	—	Straight line *	Straight line *
Cone	—	—	—	—	—	—	—	Straight line *

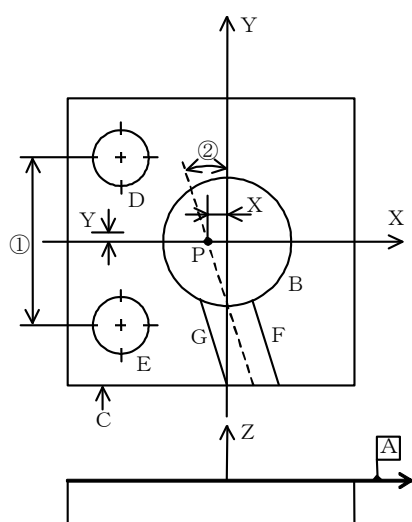
×: Inappropriate combination

— : Same as the opposite combination

** The center points of circle, oval, and sphere are calculated as point elements.

* The center axis of cylinder and cone is calculated as a straight line element.

Example of symmetry element: 1) Circle and circle, 2) straight line and straight line

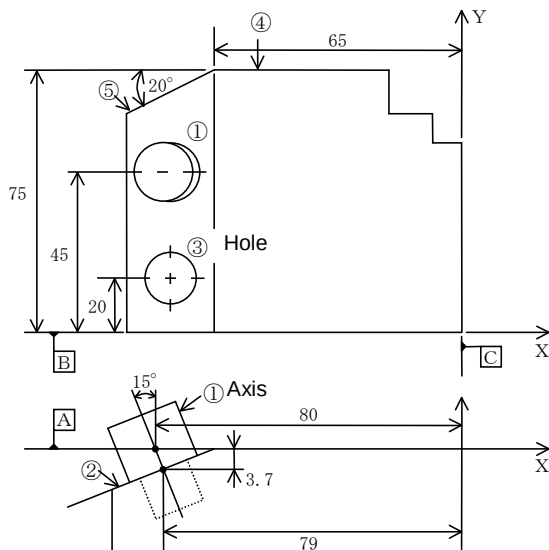


		A				
		B				
		C				
1)		D		E		Distance = ① X= Y=
2)		F		G		X= AVY=②

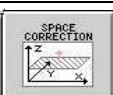
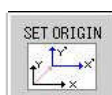
* For measuring a straight line, F and G are measured in the same direction.

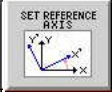
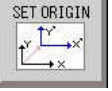
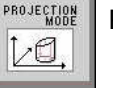
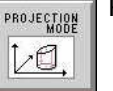
Measurement procedure

12. Exercise 4



Reference flat surface: Surface A
 Origin point specification: Edge B and Edge surface C
 Reference axis specification: Edge B

Address	Icon and operation method	Remarks	Nominal value
	 		
	   Probe number: 0 ENTER At least four points in the ball Origin point block is measured and press Terminate. Probe number: input "0", probe diameter: input "_", standard deviation: input "_". Is it accepted? (0: No, 1:Yes): Input "1" and press ENTER .	The probe to be used for setting the first coordinate system and the measurement is registered.	
A0001	 Measure at least three points on the Surface A, and press Terminate.		
C0001 C0002	  	Setting the surface A as the reference surface (XY surface). The perpendicular line of Surface A is parallel to the Z axis. On the Surface A becomes Z=0.	
A0002	  Measure at least two points on the Edge B, and press Terminate.		

Address	Icon and operation method	Remarks	Nominal value
C0003 C0004	  	X axis is set in parallel to the Edge B. On the Edge B becomes Y=0.	
A0003 C0005	   	Measure at least one point on the Edge surface C, and press Terminate. Only for the X axis direction, the Origin point shifts in parallel.	
A0004	  Measure cylinder ① at least six points, and press Terminate. For each of the 1st to the 3rd points and the 4th to the 6th points, however, make a measurement so as to form a flat surface perpendicular to the cylindrical axis as much as possible. For the 7th and the latter points, any location is accepted for the measurement.	The intersection of the cylindrical axis and the measurement flat surface is obtained.	Diameter =14 X= -80 Y=45.0 AVZ=15°
A0005	 Measure at least three points on the Flat surface ②, and press Terminate.		
C0006	 	The flat surface ② is set as a slope surface.	
	   Projection mode: 3 ENTER	The projection surface is set to a slope surface.	
A0006	  Measure at least three points on the Hole ③, and press Terminate.	The middle point of the Hole ③ on the Flat surface ② is obtained.	Diameter =10 X= -79 Y=20 Z= -3.7
	   Projection mode: 1 ENTER	The projection mode is turned ON.	

Address	Icon and operation method	Remarks	Nominal value
A0007	 Measure at least two points on the Edge ④, and press Terminate.		
A0008	 Measure at least two points on the Edge ⑤, and press Terminate.	In a similar case, the icon specification is not required because the straight line measurement is executed immediately before.	
A0009	 Element address number: [-2.-1]: ENTER	The intersection and the crossing angle between the Edge ④ and ⑤ are obtained.	Crossing angle =20° X= -65 Y=75

A0001	A05	PLN	Z	-8.108				
			AVXY	-120.59.41	AVZ	0.04.45		
C0001	A80	COD2	Reference surface set					
C0002	A81	COD2	Origin point set					
A0002	A01	LIN	Y	-314.973				
			AVXY	-0.29.46				
C0003	A82	COD2	Reference axis set					
C0004	A81	COD2	Origin point set					
A0003	A00	PNT	X	-214.328				
C0005	A81	COD2	Origin point set					
A0004	A06	CYL	Diameter	13.980	Radius	6.990		
			X	-79.828	Y	45.351		
			AVXY	-179.33.41	AVZ	14.38.19		
A0005	A05	PLN	Z	17.326				
			AVXY	-179.57.43	AVZ	14.52.32		
C0006	A83	COD3	Slope surface set					
A0006	A02	CIR	Diameter	10.198	Radius	5.099		
			X	-78.931	Y	20.117	Z	-3.636
A0007	A01	LIN	Y	74.963				
			AVXY	179.59.51				
A0008	A01	LIN	Y	98.435				
			AVXY	-160.12.53				
A0009	A61	PNT	Crossing angle	19.47.16				
			X	-65.241	Y	74.996	Z	0.000

Explanation of exercise 4

〈 Cylinder measurement (A06) 〉

Measure it for 6 points or more but 200 points or less, and press Terminate.

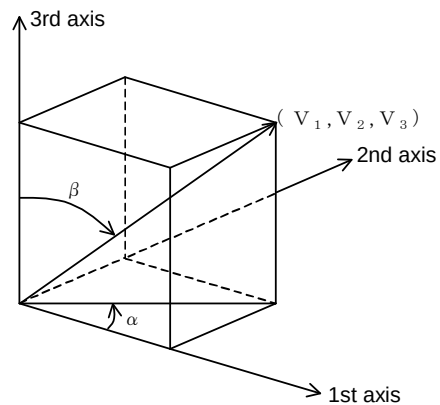
For the first three points and the following three points, make a measurement on the circumference perpendicular to the cylindrical axis as much as possible.

For the 7th and the latter points, measure any location.

Take long distance as much as possible between the circle center formed by the first three points and the circle center formed by the following three points.

Output items:... Diameter, radius, intersection between cylindrical axis and measurement surface.

When measuring seven points or more for the direction vector of cylindrical axis: cylindricity, standard deviation, maximum and minimum values are also output.



$(V_1, V_2, V_3) = \text{Direction vector}$

$$V_1 = \cos(\alpha) * \sin(\beta)$$

$$V_2 = \sin(\alpha) * \sin(\beta)$$

$$V_3 = \cos(\beta)$$

α : The angle to the 1st axis when projecting the direction vector on the measurement surface (flat surface angle)

β : The slant angle from the 3rd axis (solid angle)

Flat surface	1st axis	2nd axis	3rd axis	α	β
XY	X	Y	Z	AVXY	AVZ
YZ	Y	Z	X	AVYZ	AVX
ZX	Z	X	Y	AVZX	AVY
YX	Y	X	Z	AVYX	AVZ
ZY	Z	Y	X	AVZY	AVX
XZ	X	Z	Y	AVXZ	AVY

< Intersection element (A61) >

The intersection elements (intersection point and intersection line) of the geometry elements stored in two address numbers immediately before are obtained. The result of the output is stored as a geometry element. The combination of the intersection elements is shown as follows.

	Point	Straight line	Circle	Oval	Sphere	Flat surface	Cylinder	Cone
Point	×	Point	×	×	×	Point	Point *	Point *
Straight line	—	Point	Point	Point	Point	Point	Point	Point
Circle	—	—	Point	Point	×	×	×	×
Oval	—	—	—	×	×	×	×	×
Sphere	—	—	—	—	×	Circle	Point *	Point *
Flat surface	—	—	—	—	—	Straight line	Point *	Point *
Cylinder	—	—	—	—	—	—	Point #	Point #
Cone	—	—	—	—	—	—	Point #	Point #

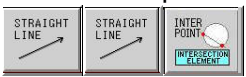
×: Inappropriate combination

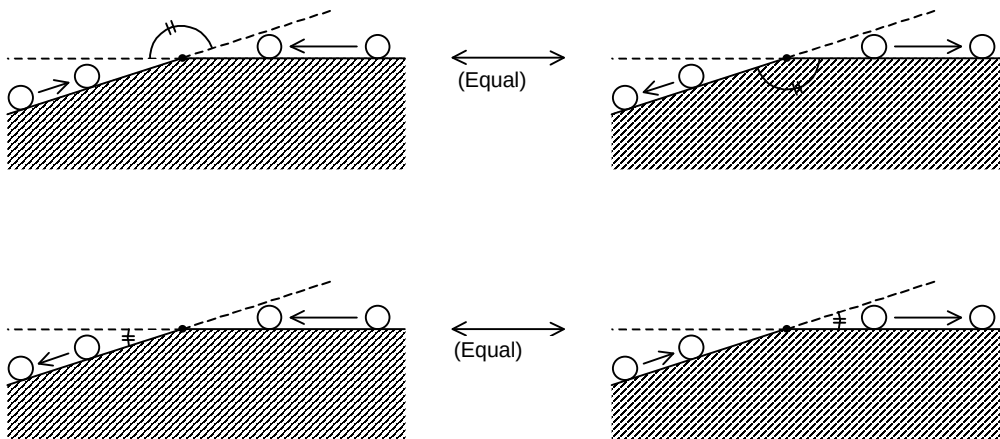
— : Same as the opposite combination

* The center axis of cylinder and cone is calculated as a straight line element.

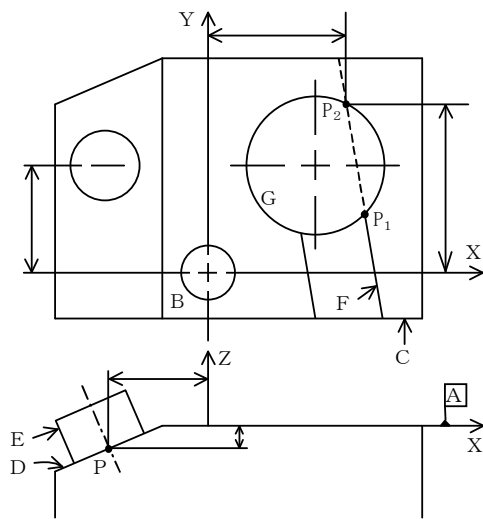
#: Surface intersection and the axis center intersection are selected.

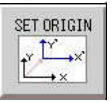
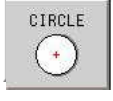
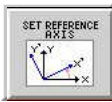
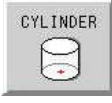
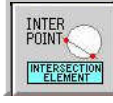
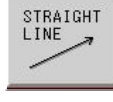
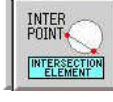
The intersection points and the intersection angle of the two edges are obtained by

the , however, depending on the straight line measurement direction, the intersection angle to be output becomes as follows.



Example of intersection element: 1) Flat surface and cylinder , 2) straight line and circle



	  
	 
	 
1)	  
2)	   The coordinate value is displayed because there are two intersection points (P1, P2). Select the P2.

MEMO

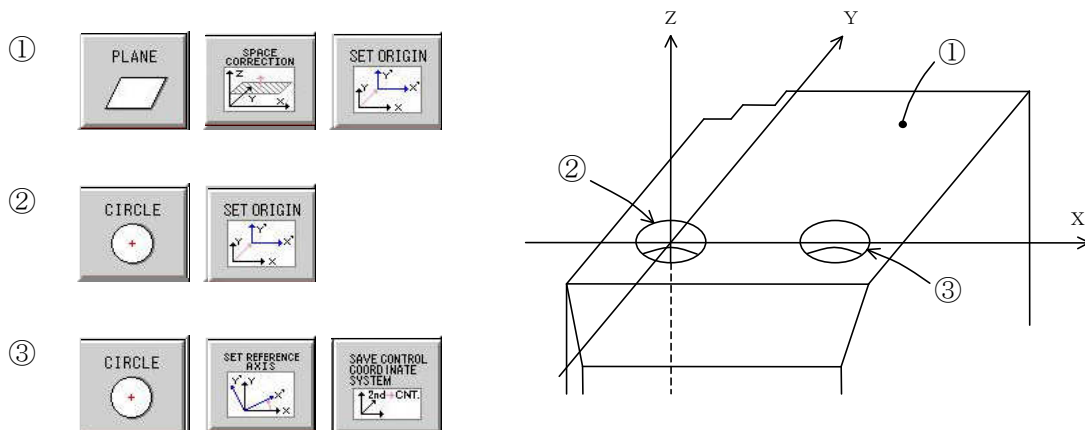
13. Teaching auto measurement

For measuring the same work two or more times, a part program is created in the teaching measurement, so that auto measurement (CNC) can be executed for the second and the subsequent works based on the program.

1) Control coordinate system setting

The setting must be made in order to set the location of the work when executing the teaching measurement and the CNC measurement.

〈 Example 〉



Then, the coordinate system as the right figure is formed and the location of the work is set. And the behavior of the CNC is controlled based on this coordinate system.

2) Start of teaching measurement



Part program name: _____ **ENTER**

The program name to be created from now is input.

Eight or less characters including the alphanumeric characters, "#", "!", "&", "\$", "+", and "-" are accepted for input.

(On the screen, it is displayed as "Teach: Program name").

〈 Procedure registration 〉

Keep on pressing the process code for the 1st work, and press the **Middle Point** ^(Note) key for registration.

To input the middle point, use the **•** key on the operation board.

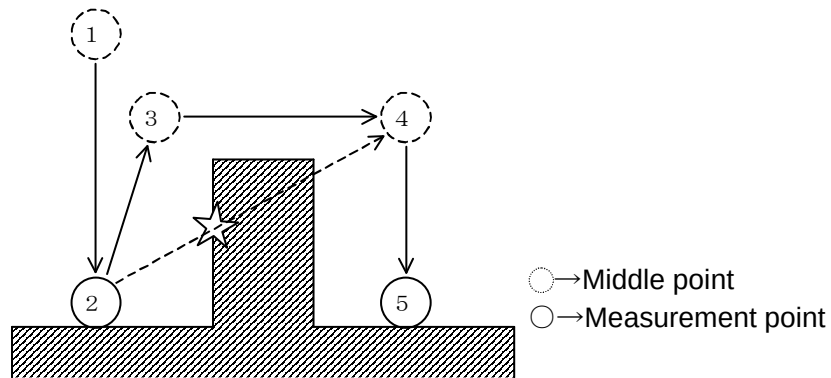
Notes

Middle point is a position where the probe shifts to (routing point for detour operation) in non measurement operation.

The next page shows an example of measuring an edge surface in which the **middle point** is pressed in ①, and then, the measurement ② is executed.

Next, the **middle point** is pressed in ③ and ④, and the measurement ⑤ is executed.

If the **middle point** is not pressed by accident in ③, it moves like the wavy line and crushes.



3) Exit of teaching measurement



(On the screen, it is redisplayed as "Manual" instead of "Teach")

4) Auto (CNC) measurement preparation

The control coordinate system is set.

Never forget to make this setting, otherwise a malfunction may be occurred. However, immediately after a teaching measurement, the setting can be skipped if the probe or the control coordinate system is not changed. For the travel speed, measurement speed, return speed, return amount, stopping position, tolerance, and passing position tolerance, the numeric values input in the system registration (SYS) are used.

In addition, the numeric values can be changed by executing a control process.

5) Starting auto (CNC) measurement



Part program name: _____ **ENTER**

Inputting the program name created by the teaching measurement

Execution stop line number: Press **ENTER**

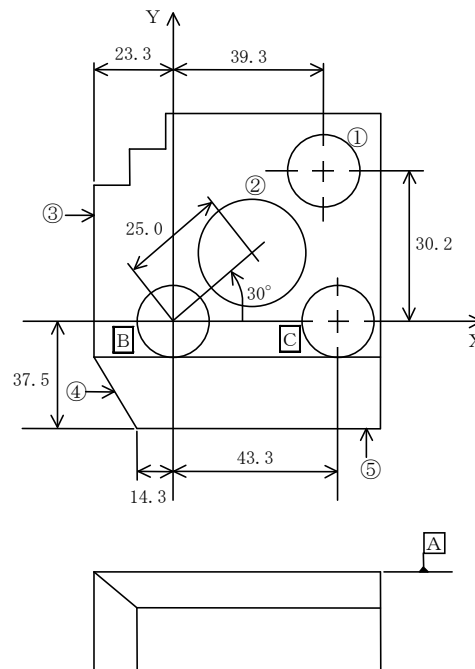
To stop the auto measurement on the way, input the part program line number at this time. By just pressing **ENTER** without inputting any value, the process is executed to the last.

(On the screen, it is displayed as "Auto: Program name").

When all the measurements are finished, the CNC measurement is terminated automatically.

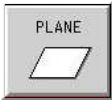
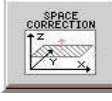
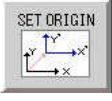
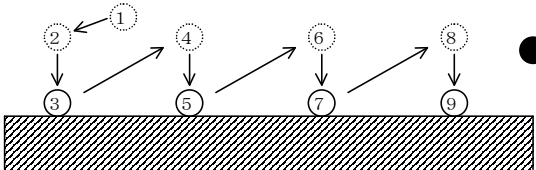
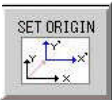
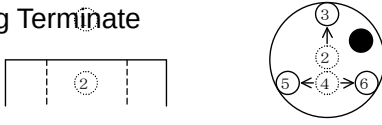
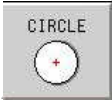
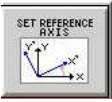
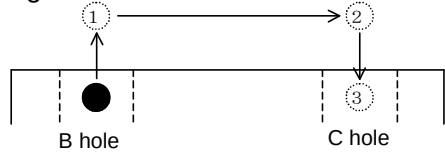
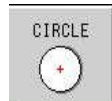
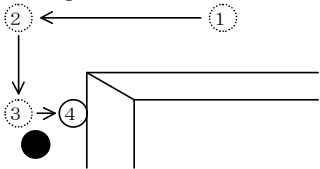
Example of teaching auto (CNC) measurement

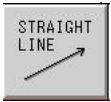
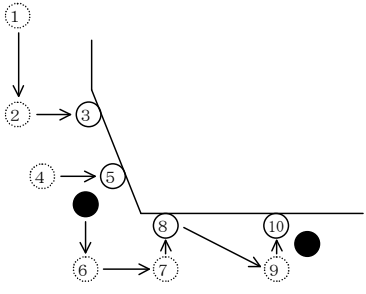
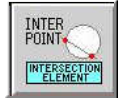
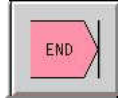
As an example of measurement, a work diagram and a procedure are shown in the figure below.



Reference flat surface: Surface A
Origin point specification: Hole B
Reference axis specification; Hole C

Address	Icon	Operation	Remarks
Preparation 1		Execution of calibration	
Preparation 2 (CNC)		Measuring at least three points on the Surface A, and pressing Terminate	Reference surface setting
		Measuring at least three points on the Hole B, and pressing Terminate	Origin point setting
		Measuring at least three points on the Hole C, and pressing Terminate	Reference axis setting
			Control coordinate setting
		Part program name: 687 ENTER	Restarting of teach startup measurement

Address	Icon	Operation	Remarks
A0001 C0001 C0002	  	Measuring at least three points on the Surface A, and pressing Terminate 	○ → Middle point ○ → Measurement point ● → Terminate
A0002 C0003	 	Measuring at least three points on the Hole B, and pressing Terminate 	Diameter=12
A0003 C0004	 	Measuring at least three points on the Hole C, and pressing Terminate 	Diameter=12
A0003		The Hole C position is displayed in the cylindrical coordinate.	Radius vector R=43.3
A0004		Measuring at least three points on the Hole ①, and pressing Terminate.	Diameter=12 X = 39.3 Y = 30.2
A0005		Measuring at least three points on the Hole ②, and pressing Terminate	Diameter=28
A0005		The Hole ② position is displayed in the cylindrical coordinate.	Radius vector R = 25 Angle = 30°
A0006		Measuring at least one point on the Edge surface ③, and pressing Terminate. 	X = -23.3

Address	Icon	Operation	Remarks
A0007		Measuring at least two points on each of the Edge ④ and ⑤, and pressing Terminate. 	
A0008			
A0009		Element address number: [-2.-1]: ENTER	$X = -14.3$ $Y = -37.5$
Move the probe adjacent to the first middle point at the time of starting the teaching, and input the last middle point.			
			Exit of teach
		Part program name: <u>687</u> ENTER Execution stop line number: ENTER	

Measurement day:00/03/14	Measurement time:10:30:56
Measured by: TOKYO SEIMITSU CO.,LTD	Supplier:
Components provided by: Test work	Model No./Drawing No.: TSK-001
Note:	

Address	Code	Label	Symbol	Actual measurement value	Nominal value	Deviation	Upper limit tolerance	Lower limit tolerance	Exceeding amount
A0001	A05	PLN	Z AVXY	-7.855 124.11.58	AVZ	0.03.55			
C0001	A80	COD2	Reference axis set (X Y)						
C0002	A81	COD2	Origin point set						
A0002	A02	CIR	Diameter X	12.038 -238.605	Y	-14.289			
C0004	A81	COD2	Origin point set						
A0003	A02	CIR	Diameter X	12.041 43.303	Y	0.272			
C0003	A82	COD2	Reference axis set (X)						
A0003	B38	CIR	<u>Radius</u> <u>vectorR</u>	<u>43.304</u>	AXY	0.00.00	Z		0.000
A0004	A02	CIR	Diameter X	12.036 39.332	Y	30.172			
A0005	A02	CIR	Diameter X	28.055 21.654	Y	12.501			
A0005	B38	CIR	Radius	25.003	AXY	29.59.52	Z		0.000
A0006	A00	PNT	<u>X</u>	<u>-23.314</u>					
A0007	A01	LIN	X AVXY	-27.886 -69.59.55					
A0008	A01	LIN	Y AVXY	-37.475 -0.00.25					
A0009	A61	PNT	Crossing angle <u>X</u>	69.59.30 <u>-14.246</u>	<u>Y</u>	<u>-37.474</u>	Z		0.000

Part program list and content



File type: (0: Part program, 1: Part placement, 2: Logging, 3:0&1) , Input "0" and press **ENTER**.

Drive number (0-17): Input "0" and press **ENTER** (Hard disk →0, Floppy disk →1).

File name: Input "687" and press **ENTER**.

↑ Input the part program name to be printed out.

〈Example of print〉

Line	SQ/SYB	Label	Code	Information		
1	Q0001		B30			
2	Q0002		F30	-13.054	-5.343	71.016
3	Q0003	PLN	A05			
4			F31	-13.053	-5.362	4.020
5			F30	32.868	-4.352	6.963
.....						
9			F32	11.552	46.614	35.651
10	Z	PLN	K07			
11	AVXY		K07			
12	AVZ		K07			
13	FLAT		K07			
14	Q0004	COD2	A80			
15	Q0005	COD2	A81			

Q0001-: Sequence NO.
 Z-: Output item
 A** - C** : Each processing code
 F30 : Middle point
 F31 : Probing point
 F32 : Terminate point
 Information column: Probe center data (X,Y,Z)

Specification of output items

The data items to be output after measurement can be specified in advance.



Specification of output items (regular basis)

(B11)

The output items to be output are always limited to the ones specified here.

In the case of a circle, as an example, the specified elements include: (1: dot, 2: straight line, 3:circle, 4: oval, 5:sphere, 6: flat surface, 7: cylinder, 8:cone, 9. width): 3

Selection code(1:DC,2:RC,3:XYZ,4:ROUND,5:SIGM) [12345]: 134

If the above setting is applied for a measurement treating a circle, the data of "DC, XYZ, and ROUND" is always supposed to be output.

14. Nominal value collation

If the Nominal value collation setting is turned on, each item is displayed in the message column on the monitor each time after a measurement is finished, and the key code input is awaited.

The key codes include nine types which are shown below.

〈Key code: 0〉

Nothing is output for the item.

〈Key code: 1〉

The actual measurement value is output for the item. The label is also input.

〈Key code: 2〉

The Nominal value collation is executed. The label and the Nominal value are input.

〈Key code: 3〉

The nominal value collation (with tolerance) is executed. The label, nominal value, upper limit tolerance, and lower limit tolerance are input.

〈Key code: 4〉

The nominal value collation (with general tolerance) is executed. The label and the nominal value are input. For the tolerance, since the general tolerance is applied, nothing is input.

〈Key code: 6〉

Everything after that item is not output.

〈Key code: 7〉

Everything after that item is output in actual measurement value.

〈Key code: 8〉

The key code input waiting state of the line immediately before is returned.

〈Key code: 9〉

The process is cancelled.

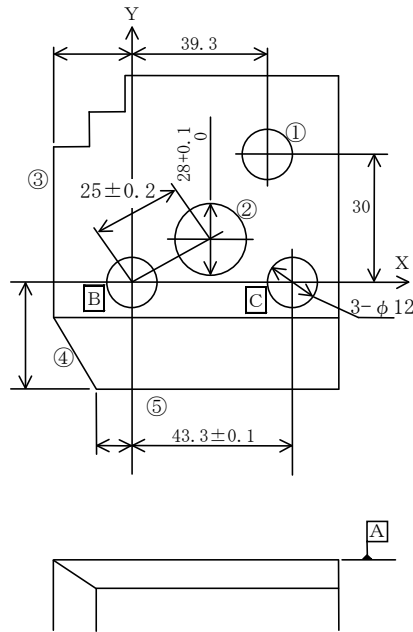
Note) Input method of nominal value

(1) = Input with code → Deviation: (Actual measurement value) - (Nominal value)

(2) = Input without code → Deviation : [(Actual measurement value)] - [(Nominal value)]

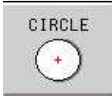
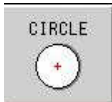
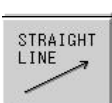
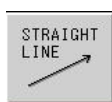
In the nominal value collation with tolerance, if the nominal value is "0", input the value as "+0".

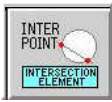
Example of nominal value (Nominal value) collation measurement



Reference flat surface: Surface A
Origin point specification: Hole B
Reference axis specification; Hole C

Address	Icon menu	Measuring method and input	Remarks
		Nominal value collation mode: 1 ENTER	Turning on the nominal value collation
A0001		Measuring at least three points on the Surface A, and pressing Terminate. Z key code: 6 ENTER	No output for entire Surface A measurement
C0001 C0002			
A0002		Measuring at least three points on the Hole B, and pressing Terminate. Diameter key code: 1 ENTER Label: CIR ENTER Radius key code: 6 ENTER	Hole B measurement Actual measurement value is output for the diameter No output for the radius and after
C0003			
A0003		Measuring at least three points on the Hole C, and pressing Terminate. Diameter key code: 1 ENTER Label: CIR ENTER Radius key code: 6 ENTER	Hole C measurement Actual measurement value is output for the diameter No output for the radius and after

Address	Icon menu	Measuring method and input	Remarks
A0003		Radius vector R: key code: 3 ENTER Label: CIR ENTER Nominal value: 43.3000 ENTER Upper limit tolerance: 0.1000 ENTER Lower limit tolerance: -0.1000 ENTER AXY key code: 6 ENTER	The Hole C is output in the cylindrical coordinate. Only the radius vector R is applied for the nominal value collation.
A0004		Measuring at least three points on the Hole ①, and pressing Terminate. Diameter key code: 1 ENTER Label: CIR ENTER Radius key code: 0 ENTER X key code: 2 ENTER Label: ENTER Nominal value: 39.3 ENTER Y key code: 2 ENTER Label: ENTER Nominal value: 30.2 ENTER	Hole ① measurement Actual measurement value is output for the diameter. No output for the radius. Nominal value collation is applied for X and Y.
A0005		Measuring at least three points on the Hole ②, and pressing Terminate. Diameter key code: 3 ENTER Label: CIR ENTER Nominal value: 28.0000 ENTER Upper limit tolerance: 0.1000 ENTER Lower limit tolerance: 0.0000 ENTER Radius key code: 6 ENTER	Hole ② measurement Actual measurement value is collation for the diameter. No output for the radius and after.
A0005		Radius vector R: key code: 3 ENTER Label: CIR ENTER Nominal value: 25.0000 ENTER Upper limit tolerance: 0.2000 ENTER Lower limit tolerance: -0.2000 ENTER AXY key code: 6 ENTER	The Hole ② is output in the cylindrical coordinate. Only the radius vector R is applied for the nominal value collation.
A0006		Measuring at least one point on the Edge surface ③, and pressing Terminate. X key code: 1 ENTER Label: PNT ENTER	Edge surface ③ measurement. Outputting X in the actual measurement value.
A0007		Measuring at least two points on the Edge ④, and pressing Terminate. X key code: 6 ENTER	No output for entire Edge ④ measurement
A0008		Measuring at least two points on the Edge ⑤, and pressing Terminate. Y key code: 6 ENTER	No output for entire Edge ⑤ measurement

Address	Icon menu	Measuring method and input	Remarks
A0009		Element address number: [-2..1]: ENTER Crossing angle key code : 0 ENTER X key code : 1 ENTER Label : PNT ENTER Y key code : 1 ENTER Label : ENTER Z key code : 0 ENTER	Intersection between the Edge ④ and ⑤. Only X and Y are output in the actual measurement value.
		Nominal value collation mode: 0 ENTER	The nominal value collation is turned OFF.

Print data after measurement

Measurement day: 00/03/14					Measurement time: 13:30:25				
Measured by: TOKYO SEIMITSU CO.,LTD					Supplier:				
Components provided by: Test work					Model No./Drawing No.:TSK-001				
Note:									
Address	Code	Label	Symbol	Actual measurement value	Nominal value	Deviation	Upper limit tolerance	Lower limit tolerance	Exceeding amount
C0001	A80	COD2	Reference surface set (XY)						
C0002	A81	COD2	Origin point set						
A0002	A02	CIR	Diameter 12.018						
C0004	A81	COD2	Origin point set						
A0003	A02	CIR	Diameter 12.001						
C0003	A82	COD2	Reference axis set (X)						
A0003	B38	CIR	Radius	43.308	43.300	0.008	0.100	-0.100	+
			Vector R						
A0004	A02	CIR	Diameter	12.036					
			X	39.332	39.300	0.032			
			Y	30.188	30.200	-0.012			
A0005	A02	CIR	Diameter	28.131	28.000	0.131	0.100	0.000	0.031
A0005	B38	CIR	Radius	24.969	25.000	-0.031	0.200	-0.200	-
			Vector R						
A0006	A00	PNT	X	-23.278	Error margin		Upper/Lower limit tolerance		
A0009	A61	PNT	X	-14.401					
			Y	-37.512					

If the deviation does not exceed the tolerance, each of the upper tolerance and the lower tolerance is divided into four, which is output with the "+" and "-" codes.
If the nominal value is 25.000, and the upper and lower tolerances are plus or minus 0.2 as the above radius value, it corresponds to the * part since the deviation is -0.031.

In the case of $25^{+0.1}_0$, there is only the + code, and the +0.1 is divided by four.

0. 2 0 0	++++
0. 1 5 0	+++
0. 1 0 0	++
0. 0 5 0	+
0. 0 0 0	
-0. 0 5 0	- (*)
-0. 1 0 0	--
-0. 1 5 0	---
-0. 2 0 0	----

Tolerance judgment result
Numerical value :
tolerance exceeding amount
+=+ Error margin
-=- Error margin

MEMO

15. Other processes concerning CNC measurement

15-1 Continuous measurement using set jig (for 1 unit)

If a set jig is provided, the measurement can be executed continuously by pausing the measurement at the final line of the procedure, replacing the workpiece, and jumping to the beginning of the procedure, since it is possible to set a workpiece at the same position in the same condition even after the workpiece is replaced.

1) Mount the 1st workpiece to the set jig to create a part program.

2)  Part program name: press **ENTER**. Teach continuation line number[]:"__"press **ENTER**.
Input the part program name created with 1) to restart the program. The final line is displayed. Input the line number for continuation.

3)  It is paused at this line.

4)  Part program name: press **ENTER**. Sequence number: input "Q1", and press **ENTER**.
Input the part program name of 1).

5)  Exit of teaching

6)  Part program name: press **ENTER**. Execution stop line: press **ENTER**.
The CNC measurement is executed until the stop line and is paused.

7) Mount the following workpiece to the set jig.

8)  The procedure is continued and is executed from the 1st line by the jump at the execution line.

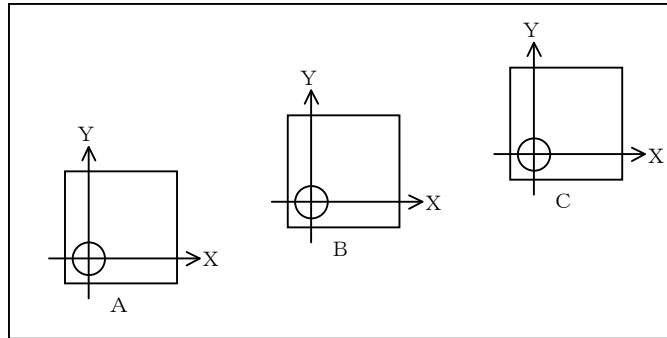
For the next and the following workpieces, repeat the steps 7) and 8).

15-2 Continuous measurement using set jig (for two or more units)

All the workpieces can be measured continuously by mounting the workpieces of the same shape to the set jig in almost parallel to the machine.

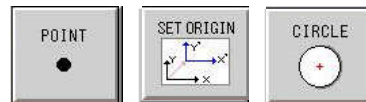
For that purpose, it is necessary to register information in the file (part placement file) about the position of the surface plate to which each workpiece is set.

Example) When continuously measuring the following workpieces in the order of A, B, and C.



- 1) Mount the A, B, and C workpieces to the set jig and create a part program with the Workpiece A. After setting the 2nd coordinate system which is currently in teaching process, execute the control coordinate system setting (A95).

- 2) The relative position of each workpiece is obtained. Measure the B and C workpieces according to the following procedure.



- 3)  Part program name:

Input the part program name created with 1).

Input coordinates value (0: Coordinates value specification, 1: Pitch specification, 2: Element address specification): input "2", and press ENTER.

To the inquiry of "Element address No. for reference workpiece", input the element address No. at the time of executing a circle measurement for the Workpiece A.

To the inquiry of "Element address No. for the 1st workpiece", input the element address No. at the time of executing a circle measurement for the Workpiece B in 2).

Similarly, input the element address number at the time of executing a circle measurement for the 2nd workpiece (C).

To exit the registration of the address No. for all the workpieces, input ":(colon)", and press or only press , to the inquiry of "Element address No. for nth workpiece".

4)



Part program name: ENTER

↓
Input the part program name created with 1).

Execution stop line: ENTER

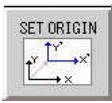
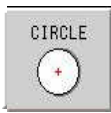
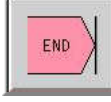
Then, continuous CNC measurement is started.

* In this example, the coordinate value in the 0th coordinate system with the A workpiece being set as the starting point is registered for the part placement file information.

15-3 Loop process (parallel shift in X direction)

Measurement procedure

Workpiece chart $X_o = 20$ $Y_o = 15$ $X_d = 25$ X_w-Y_w : Control coordinate system X_o, Y_o : Offset input value X_d : Interval input value			
Address	Icon and operation method	Remarks	Nominal value
	Since a part program is created by executing a teaching measurement, set the Y_w-Y_w coordinate of the above figure, and press in advance.		
	Part program name:____ ENTER	Start of teaching measurement (making a measurement while inputting middle points)	
		Restarting measurement	
A0001	Measuring at least three points on the Surface A, and pressing Terminate	Setting of reference surface	
C0001 C0002			
A0002	Measuring at least two points on the Edge B, and pressing Terminate	Setting of reference axis and the starting point in Y axis direction	
C0003 C0004			
A0003	Measuring at least one point on the Edge C, and pressing Terminate		

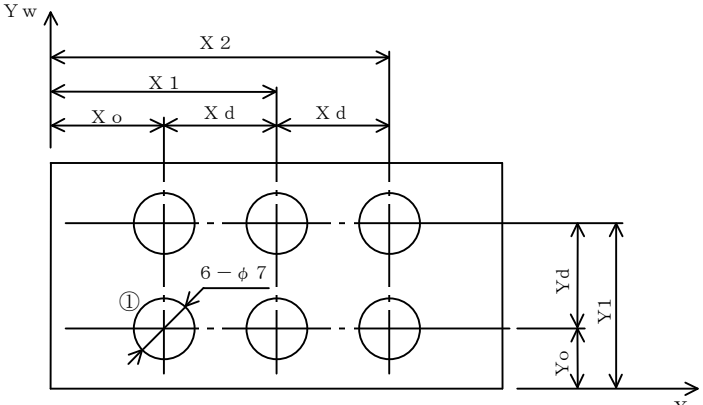
Address	Icon and operation method	Remarks	Nominal value
C0005 C0006	 	Setting of starting point in X axis direction Control coordinate system setting	
	 Loop type: 0 ENTER Reference flat surface: 1 ENTER Increment: 25,0,0 ENTER Initial position: 20,15,0 ENTER Loop frequency: 3 ENTER	Beginning of repetition Inputting the interval in X axis direction Inputting the 1st offset position Inputting the number of repetition	
A0004	 Pressing the middle point at the upper part of the Hole ①, measuring at least three points, and pressing Terminate Returning to the upper part of the Hole ① and pressing middle point		Diameter =10 X =20 Y =15
		End of repetition	
		Exit of teaching measurement	
	 Part program name: ENTER Execution stop line: ENTER	Start of the CNC measurement Measuring three holes continuously	

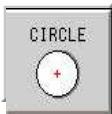
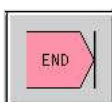
Note 1) The loop process is a measurement that repeats the same pattern.
 It shifts the control coordinate system and repeats the measurement.
 Due to the loop end (C04), it returns to the initial control coordinate system automatically.
 However, if interruption process or cancellation is executed, return to the initial control coordinate system by executing the coordinate system recall (A90) and the control coordinate system setting (A95), since the control coordinate system is shifting.

Note 2) Never fail to set the control coordinate system after setting the 2nd coordinate system in view of executing the Nominal value collation.

15-4 Loop process (parallel shift in X and Y directions)

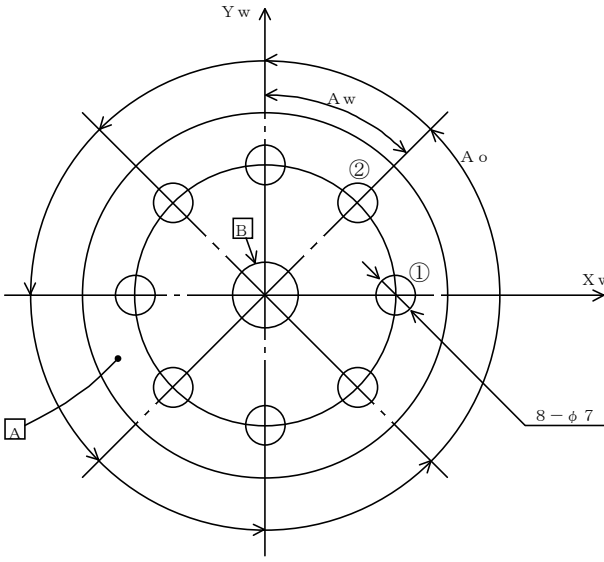
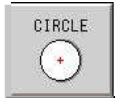
Measurement procedure

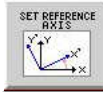
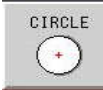
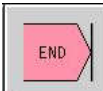
Workpiece chart  $X_0 = 15$ $Y_0 = 10$ $X_d = 25$ $Y_d = 20$ X_w-Y_w : Control coordinate system X_0, Y_0 : Offset input value X_d, Y_d : Interval input value			
Address	Icon and operation method	Remarks	Nominal value
	Since a part program is created by executing a teaching measurement, set the X_w-Y_w coordinate of the above figure, and press  in advance.		
	Executing the * part of P57 and 58		
	 Loop type: <u>0</u> ENTER Reference flat surface: <u>1</u> ENTER Increment: <u>0.20,0</u> ENTER Initial position: <u>15,10,0</u> ENTER Loop frequency: <u>2</u> ENTER	Beginning of repetition (Y direction) Inputting the interval in Y axis direction Inputting the 1st offset position Inputting the number of repetition	
	 Loop type: <u>0</u> ENTER Reference flat surface: <u>1</u> ENTER Increment: <u>25,0,0</u> ENTER Initial position: <u>0,0,0</u> ENTER Loop frequency: <u>3</u> ENTER	Beginning of repetition (X direction) Inputting the interval in X axis direction Inputting the 1st offset position Inputting the number of repetition	

Address	Icon and operation method	Remarks	Nominal value
A0004	 Inputting the middle point at the upper part of the Hole ①, measuring at least three points, and pressing Terminate. Returning to the upper part of the Hole ① and pressing middle point		Diameter =10 X=15 Y=10
		End of repetition in X direction	
		End of repetition in Y direction	
		Exit of teaching measurement	
	 Part program name: _____ ENTER Execution stop line: ENTER	Start of the CNC measurement Six holes are continuously measured	

15-5 Loop process (rotation by angle)

Measurement procedure

Workpiece chart  $A_o = 45$ $A_w = 45$ X_w-Y_w : Control coordinate system A_o : Offset input value A_w : Interval input value			
Address	Icon and operation method	Remarks	Nominal value
	Since a part program is created by executing a teaching measurement, set the X_w-Y_w coordinate of the above figure, and press  in advance.		
	 Part program name: ENTER	Start of teaching measurement (making a measurement while inputting middle points)	
		Restarting measurement	
A0001	 Measuring at least three points on the Surface A, and pressing Terminate	Setting of reference surface	
C0001 C0002	 		
A0002	 Measuring at least three points on the Hole B, and pressing Terminate	Setting of starting point	
C0003			

Address	Icon and operation method	Remarks	Nominal value
A0003 C0004 C0005	Continuously measuring at least three points on the Hole ①, and pressing Terminate  	Setting of reference axis Control coordinate system setting	Diameter = 7
C0004 C0005	 1 ENTER	Changing the data output format in cylinder	
	 Loop type: 2 ENTER Reference flat surface: 1 ENTER Increment: 0,45,0 ENTER Initial position: 25,45,0 ENTER Loop frequency: 7 ENTER	Beginning of repetition Inputting the interval in angle direction Inputting the Hole ② position Inputting the number of repetition	
A0004	 Inputting the middle point at the upper part of the Hole ②, measuring at least three points, and pressing Terminate. Returning to the upper part of the Hole ② and pressing middle point		Diameter=7 Radius vector R:25 AVXY:45°
		End of repetition	
		Exit of teaching measurement	
	 Part program name: ENTER Execution stop line ENTER	Start of the CNC measurement 8 holes are continuously measured	

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16. Edit of part program

16-1 Editing by multi-purpose measurement program (part program mask)

The measurement is executed by skipping the lines masked by the part program mask (B76) (This process can be executed for the part program and the part placement file).

Example) For masking 10th line to 20th line of the part program name "A".

PART PG MASK	File type (0: Part program, 1: Part placement file): 0
	File name: A
	Mask process (0:Setting, 1:Cancel): Input "0", and press .
	Line number (*,*, ...*.*): input "10.20", and press .

By executing the print (B74) of the part program list, the head of the line set with the mask is appended with "*".

*To cancel the mask of the part program, input "1", and press (Cancel) to the inquiry of "Mask process".

Note) Set this process while taking care of the middle point information (F30) before and after the mask, and the probe No. (F22).

16-2 Editing by multi-purpose measurement program (edit mode)

It is executed for inserting a measurement or a process in a part program created by the teaching measurement (This process can be executed for part programs).

Example) For inserting a circle measurement after the 10th line of the part program name "A".

- 1)  Part program name: Input "A", and press **ENTER**.
Execution stop line: Input "11", and press **ENTER** (inputting the line number intended to insert).

It shifts to the Edit mode, and the CNC measurement is started.

*When the edit is started, the part PG display (B67) automatically turns "ON".

- 2) The CNC measurement is stopped at the 11th line (execution stop line) (shifting to the edit stop).
Execute the circle measurement intended to insert by applying the same operation of the teaching measurement.
- 3) Copying the procedure of the 10th line and after (There are two methods).

Method 1 (1)



- (2) Execution stop line: press ENTER (The CNC measurement is started until the final line).
- (3) Will you leave the content changed in the file: Input "1", and press ENTER.
- (4) Now, input of "output file name [A]" is awaited. If the same file name (A) is input here, the A file before the edit is erased, and the procedure after the edit is changed in the A file.
If a new file name (B) is input, the A file before the edit is left as it is, and the procedure after the edit becomes the B file.

* The method 1 is used for copying the procedure while executing a CNC measurement.

Method 2 (1)



- (2) Copy end line number[final line]: Press **ENTER**.
- (3) Input the output file name [A] (Refer to ④ of the method 1).

The procedure from the 11th line to the final line is displayed and the copying is finished.

* The method 2 is used for copying without executing the CNC measurement.
By applying this process, the edit time can be shortened.

* When the edit is finished, the part PG display (B67) automatically turns "OFF."

16-3 Edit by edit program

This process is executed for editing the already created procedures including correction and deletion. This process can be executed for part programs, parts placement files, and logging files.

1) Starting the Edit program.

- (1) Exit the Multi-purpose measurement program, and press **TOOL** on the monitor while "Execution program name:" is stated.
- (2) Select the "File edit" of the article No.6.
- (3) Select the "Multi-purpose program file" of the article No.2.
- (4) Select the "Part program" of the article No.1.
- (5) Set the disk device No. and the file name, and execute the program.

- The line intended to edit (referred to as current line) is distinguished from the other lines. The line number of the current line is displayed in the lower right of the screen. The following methods are available for shifting the current line.

	:Moves upward line by line.
 or ENTER	:Moves downward line by line.
Previous page	:Moves upward by 10 lines.
Next page	:Moves downward by 10 lines.
Inputting <u>line number</u> and pressing ENTER : Moves to the line input.	
Inputting " <u>0</u> ", and pressing ENTER : Moves to the one after the final line.	

- Various edit functions are allocated to the function keys. There are two menus (menu 1 and 2), which can be changed over by pressing the **Menu switch** on the monitor.

2) Editing.

Example: For changing the tolerance "plus or minus 0.2" of the 10th line for the part program name "A" into "plus or minus 0.1".

- (1) The current line is shifted to the 10th line.
- (2) Press the **Space** key (line correction).The current line (10th line) is displayed in the edit area (lower part of the screen).
- (3) Press  key, move the cursor "|" to "2" of the tolerance "0.200", and input "1".
- (4) Similarly, press  key, move the cursor "|" to "2" of the tolerance "-0.200", and input "1", and press **ENTER**.

3) Exit the edit program.

- (1) When the function menu 2 is displayed on the screen, press **Normal Exit**.
- (2) Select the article No.0, and press **ENTER** (exit of tool program).
Now, the input of "execution program name" is awaited.

Other function key holds the following roles.

1) Function menu 1

Inserting a line

Example) Inserting a line before the 10th line

- (1) The current line is shifted to the 10th line.
 - (2) Press Inserting Line
The current line (10th line) and "I" are displayed in the edit area (lower part of the screen). Input symbols and others, and press ENTER.
 - (3) After the input, the insertion of the 11th line is awaited (It is possible to insert continuously).
 - (4) To exit the line insertion, press ESC key.
-

Deletion and storage of line

Example) Deleting from the 10th line to the 20th line

- (1) The current line is shifted to the 10th line.
- (2) Press Deletion and storage of lines (It is displayed as "To what extent?").
Press ↓ key until the 20th line (the last line to be deleted).
From the 10th line to the 20th line become the current line. Press ENTER.

* Up to 1000 lines can be deleted at a time. The deleted lines are retained, meaning a deleted line can be restored to any line, which can be used for relocation or copy of the lines.

Restoration of lines

Example 1) Shifting the 10th line to the 20th line

- (1) The current line is shifted to the 10th line.
- (2) Press Deletion and storage of lines (It is displayed as "To what extent?").
Press ENTER (The 10th line is deleted but retained).
- (3) The current line is shifted to the 20th line.
- (4) Press Line Restoration (The deleted 10th line is restored at the 20th line).

Example 2) Copying the 10th line to the 20th line

- (1) The current line is shifted to the 10th line.
- (2) Press **Deletion and storage of lines** (It is displayed as "To what extent?").
Press **ENTER** (The 10th line is deleted but retained).
- (3) Press **Line Restoration** (The deleted 10th line is restored at the 10th line).
- (4) The current line is shifted to the 20th line.
- (5) Press **Line Restoration** (The 10th line in memory is restored at the 20th line).
* The line can be restored as many times as possible until another line is deleted and stored.

Screen hard copy

The monitor screen currently displayed is sent to the printer to produce a hard copy.

Searching character strings

Example) Searching a character string for which A05 (flat surface measurement) is executed.

- (1) The current line is shifted to the 1st line (The current line specified and the following lines become the target for the character string search. Usually, specify the 1st line).
- (2) Press Character String Search.
Character string to be searched: Input "A05", and press **ENTER**.
Column range (start and end) [total range]: press **ENTER** (total range) .
(The column range signifies the number of characters in horizontal direction in each line. By the scale on the upper part of the screen, the range extends from 1 to 80. The column range specified becomes the target for the character string search. In this example, it is possible to input as "14", and "16" and press **ENTER** for only the column range of the A05 process code).
The current line is shifted to the first line that executes A05.

Re-research

It is used for re-researching character strings.

The character strings to be re-researched includes the character strings being researched last and the column range.

Example) If **re-research** is executed after executing the **character string research**, the current line shifts to the next line of the line that executes A05.

- * The character string can be re-researched as many times as possible until another character string is researched (It is convenient when researching over all the lines).

Rewriting of character string

Example) For rewriting the character string that executes A64 (distance between two points) into A65 (coordinate difference).

(1) The current line is shifted to the 1st line (The current line specified and the following lines become the target for the character string rewriting).

(2) Press Rewriting of character string.

Character string to be replaced: input "A64", and press ENTER.

Character string to replace: Input "A65", and press ENTER.

Column range (start and end) [total range]: and press ENTER (total range).

The character string that executes A64 is rewritten and changed into A65.

The current line is shifted to the line that executed rewriting at the last.

2) Function menu 2

Reading of data

It is used for reading (inserting) part of data in a certain file.

Example) For inserting from 30th line to the 40th line of B file into the 10th line of A file

(1) From the A file, start the Edit program.

(2) The current line is shifted to the 10th line.

(3) Press Reading of data

Input file name: Input B, and press ENTER.

The range to be input (start line, final line)[all files]: Input "30" and "40", and press ENTER (for inserting all the lines of the file, press ENTER).

* For reading data, it is also possible to specify an editing file (in this example: A file).

Writing of data

It is used for writing (overwriting) part of data in a certain file into another file.

Example) Overwriting from the 10th line to the 20th line of A file into B file

(1) From the A file, start the Edit program.

(2) Press **Writing of data**.

Output file name: Input "B" and press **ENTER**.

The range to be output (start line, final line)[all files]: Input "10" and "20", and press **ENTER** (for overwriting all the lines of the file, just press **ENTER**).

* When the file name to be output already exists, a message "The specified file already exists. Do you overwrite it? 0: No, 1: Yes" is displayed.

Inputting "0", and pressing **ENTER** or just pressing **ENTER** stops the data writing.

Inputting "1", and pressing **ENTER** executes the data writing.

For the data writing, it is not possible to specify a file currently edited (in this example: A file).

Forcible termination

It is used to finish the edit of file compulsory.

* When the compulsory termination is executed, the process executed during the file edit becomes invalid, returning to the state before the edit.

However, the data writing becomes valid.

MEMO

17. User code

User code is a process code for which user can register by user own, and the frequently used measurement procedure can be registered in one code number (1-100).

Example of operation from registration to execution

1. Press  → It is displayed as "User code [1-100]:=".

By inputting the registration No. and pressing **ENTER**, it is displayed as "1st process code:".

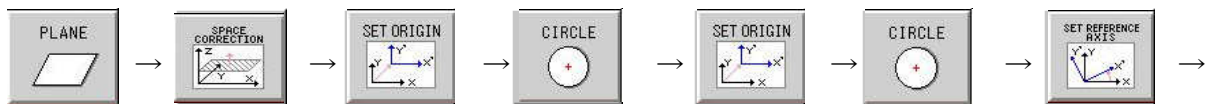
(The number signifies a registration number and up to 100 units can be registered.)

Here, input the process codes.

Up to 30 units of process codes can be input.

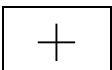
To exit the registration, press **ENTER**.

2. Input of procedure (user code [1-100]:example of registration of 5)



→ **ENTER** (end)

3. Execution of user code

By pressing  on the monitor, it is displayed as "user code:=".

By inputting a number and pressing **ENTER**, it is displayed as "user code" on the upper part of the screen. As it becomes the state of flat surface measurement and the measurement is started.

When the flat surface measurement is finished, space correction and starting point setting are executed automatically, becoming the state of next circle measurement.

As above, if the measurements are proceeded in the order of registration, and when all the measurements are finished, the user code process is finished.

-  The user code process in running is interrupted.

-  The list of user codes registered can be printed out as follows.

*** User code printout ***

CODE NO. 5

A05 A80 A81 A02 A81 A02 A82 A02

CODE NO. 10

A05 A80 A81 A02 A81 A01 A82 A02 A02 A02

CODE NO. 11

A05 A80 A81 A02 A81 A02 A82 A02 A01 A00

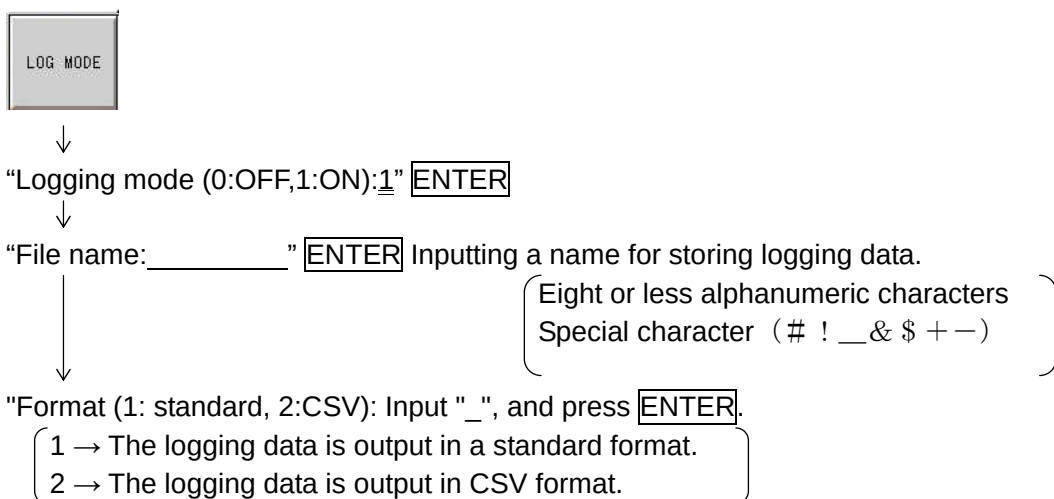
- While the execution of a user code is in process, function menus and other processes can be executed.
- While a teaching measurement is in process, a user code can be executed.
- For inputting a procedure, if wrong process code is input, exit the registration by pressing **ENTER**, and register the user code with the same registration number.

18. Logging measurement

Logging is a function that stores printed out measurement data in a disk when a workpiece is measured. By the printout function, the measurement data obtained by a measurement can be printed out again, and by the edit function, the data can be corrected or deleted.

Operation of logging measurement

1. Start of logging



Now, the logging is ready to begin, however, if the file name input already exists, the following message is displayed.

"The file exists (0: Add, 1:New):=".

0 → The output data is added from the final line of the logging file.
1 → The logging file is erased and a new logging file is created.

2. Measurement (teaching and automatic measurement are also available) <refer to P.69>.

3. End of logging

By pressing , it is displayed as "Logging mode (0: OFF, 1: ON): =". By inputting "0" and pressing **ENTER**, the logging is terminated.

To print out the above measurement data again



"Print type (0: Part program, 1: Part placement, 2: Logging, 3:0&1.):" Input "2", and press **ENTER**.



"Do you want to output Kanji (0: Yes, 1: No): Input 1", and press **ENTER**.



"Drive number (0-17): Input 0", and press **ENTER**.



"File name: Input _____", and press **ENTER**.

↑
The file name input at the time of starting logging

```
A0001 A05 PLN Z -8.067
      AVXY -31.07.31 AVZ 0.03.14
```

```
C0001 A80 COD2 Reference surface set (XY)
C0002 A81 COD2 Origin point set
```

```
A0002 A02 CIR Diameter 12.006 Radius 6.003
      X -239.641 Y -262.743
```

```
C0003 A81 COD2 Origin point set
```

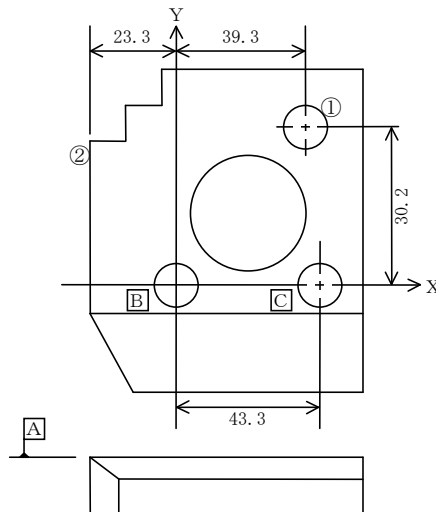
```
A0003 A02 CIR Diameter 12.004 Radius 6.002
      X 43.296 Y -0.661
```

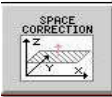
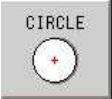
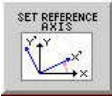
```
C0004 A82 COD2 Reference axis set (X)
```

```
A0004 A02 CIR Diameter 12.013 Radius 6.006
      X 39.331 Y 30.157
```

```
A0005 A00 PNT X -23.360
```

Example of logging measurement



Address	Icon and operation method	Remarks	Nominal value
	  Logging mode(0:OFF,1:ON): <u>1</u> File name: <u>A-1</u> Format (1:Standard, 2:CSV): <u>1</u>	Start of logging	
A0001 C0001 C0002	 Measure at least three points on the surface A, and press terminate.  	Reference surface specification	
A0002 C0003	 Measure at least three points on the Hole B, and press terminate. 	Specifying starting point	Diameter=12
A0003 C0004	 Measure at least three points on the Hole C, and press terminate. 	Reference axis specification	Diameter=12 Radius vector R=43.3
A0004	 Measure at least three points on the Hole ①, and press terminate.		Diameter=12 X = 39.3 Y = 30.2
A0005	 Measure at least one point on the Edge surface ②, and press Terminate.		X = - 23.3
	 Logging mode (0: OFF,1: ON) :0	End of logging	

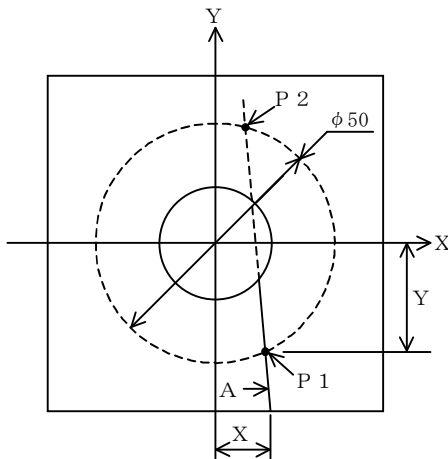
MEMO

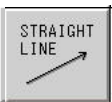
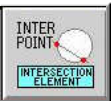
19. Key input of elements

Instead of manual input for measurement and measurement points, elements can be input using keys. The elements that can be input include: point, straight line, circle, oval, sphere, flat surface, cylinder, and cone.

Example of input

1) Intersection between circle and edge



Process code	Position	Measurement method
		Specified elements (1: point, 2: straight line, 3: circle, 4: oval, 5: sphere, 6: flat surface, 7: cylinder, 8: cone): Input "3", and press ENTER. Geometrical feature (0: inside, 1: outside, 2: no distinction between inside and outside): Input "2", and press ENTER. Input coordinate value (0: rectangular coordinate, 1: cylindrical coordinate, 2: polar coordinate): Input "[0]: ", and press ENTER. Measurement flat surface (1: XY, 2: YZ, 3: XZ): Input "1", and press ENTER. Coordinate value (X, Y): Input "0", and "0", and press ENTER. Diameter: Input "50" and press ENTER.
	Edge A	Measuring at least two points, and pressing Terminate.
		Selection of two points including P₁ and P₂ is awaited. Input "1" here.

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 27.969
X -348.406 Y -117.676

C0003 A81 COD2 Origin point set

A0003 A01 LIN Y -30.099
AVXY 3.454

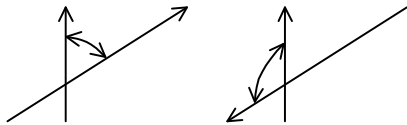
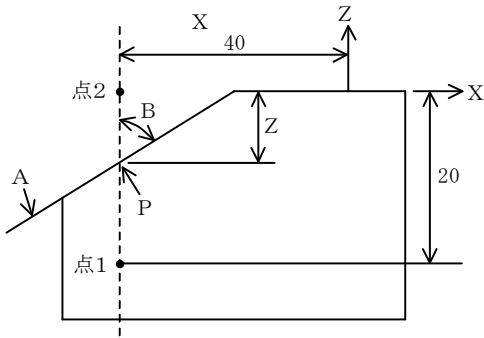
C0004 A82 COD2 Reference axis set (X)

A0004 B82 CIR Diameter 50.000
X 0.000 Y 0.000 Z 0.000

A0005 A01 LIN X 6.061
AVXY 99.943

A0006 A61 PNT X 10.072 Y -22.881 Z 0.000

2) Intersection between straight line and edge



The angle to be printed out differs according to the direction of measurement.

Process code	Position	Measurement method
ENTER ELEMENT BY KEY		Specified element: Input "2", and press ENTER (straight line). Input coordinates value: Input "0", and press ENTER . rectangular coordinate value Coordinates value 1(X,Y,Z):-40,0,-20 Coordinates value 2(X,Y,Z):-40,0,0
STRAIGHT LINE	Edge A	Measuring at least two points, and pressing Terminate
INTER POINT		The intersection P is obtained and the crossing angle (B), X, Y, and Z are printed out.

```
A0001 A05 PLN Z -88.085
          AVXY 9.31.33 AVZ 0.00.23
```

```
C0001 A80 COD2 Reference surface set (XY)
```

```
C0002 A81 COD2 Origin point set
```

```
A0002 A02 CIR Diameter 28.033 Radius 14.017
          X -286.145 Y -162.602
```

```
C0003 A81 COD2 Origin point set
```

```
A0003 A01 LIN Y -30.166
          AVXY 0.02.51
```

```
C0004 A82 COD2 Reference axis set (X)
```

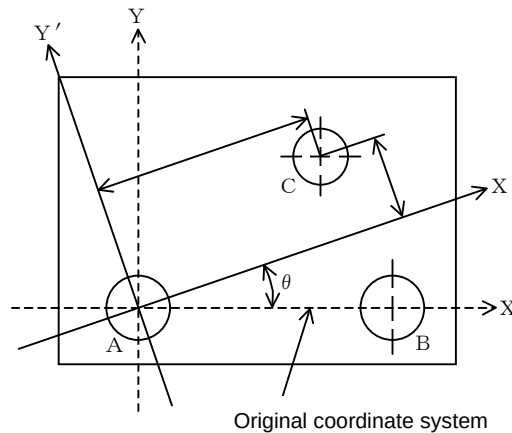
```
A0004 B82 LIN X -40.000 Y 0.000
          AVXY 110.27.54 AVZ 0.00.00
```

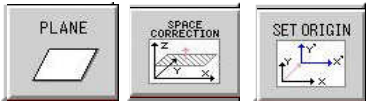
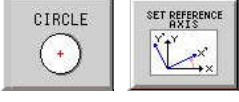
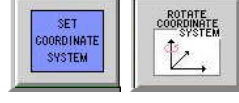
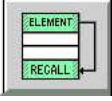
```
A0005 A01 LIN Z 6.595
          AVZX 74.48.58
```

```
A0006 A61 PNT Crossing 74.48.58
          angle
          X -40.000 Y 0.000 Z -4.261
```

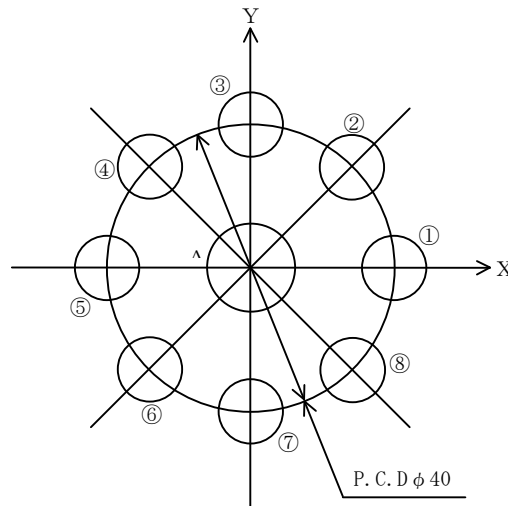
20. Example of calling measurement point

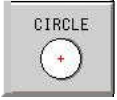
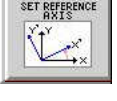
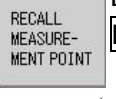
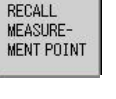
1)



Address	Process code and operation method	Position	Measurement method
A0001 C0001 C0002		Reference surface	Measuring at least three points, and pressing Terminate
A0002 C0003		Hole A	Measuring at least three points, and pressing Terminate
A0003 C0004		Hole B	Measuring at least three points, and pressing Terminate
A0004		Hole C	Measuring at least three points, and pressing Terminate
C0005	 Flat surface specification (1:XY, 2:YZ, 3:ZX, 4:YX, 5:ZY, 6:XZ) : 1 ENTER Angle*: θ ENTER		Regardless of the unit of the angles, the following shows a method of inputting angles. (1) Input by degree, minute, and second: Input "10.30.00" and pressing ENTER (2) Input by degree: Input "10.5", and press ENTER
A0005	 Element address number (Positive: absolute specification, Negative: relative specification)* : A4 ENTER (Or, A-1 ENTER)		* Input "4" for the absolute address, and input "-1" for the relative address.

2)

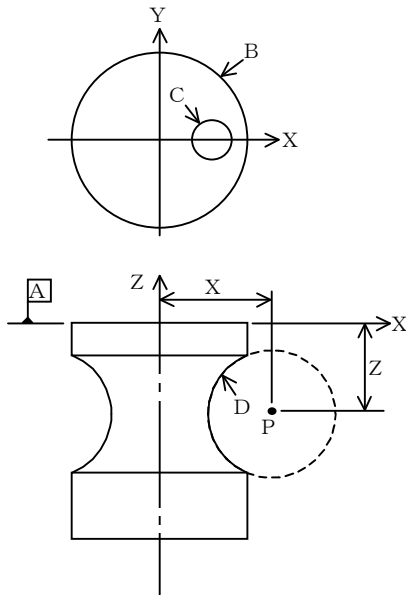


Address	Process code and operation method	Position	Measurement method
A0001 C0001 C0002	  	Reference surface	Measuring at least three points, and pressing Terminate
A0002 C0003	 	Hole A	Measuring at least three points, and pressing Terminate
A0003 C0004	 (No specification is accepted) 	Hole ①	Measuring at least three points, and pressing Terminate
A0004 ~ A00010	 (No specification is accepted)	Hole ②~ Hole ⑧	Measuring at least three points, and pressing Terminate
	 (No specification is accepted) or -1.-8   Element address number: <u>3.10</u> ENTER   Element address number:-1 ENTER		If the diameter output becomes smaller or larger for the size of the probe diameter, change the <u>dummy escape volume</u> to "roundness output x 1/2" or more, and once again start from the measurement point recall.
			Direction(0: angle direction, 1: straight line direction): <u>0</u> ENTER Standard (0: starting point, 1: measurement point immediately before): <u>0</u> ENTER Angle pitch nominal value: <u>45</u> ENTER Radius vector nominal value: <u>20</u> ENTER Address (start, end): <u>3,10</u> ENTER

21. Rotation projection mode

For the process of treating the shape of a rotating body, the measurement point is rotated and projected on a specified flat surface with the second axis working as a rotation axis.

1)



Process code	Position	Measurement method
	Edge A	Measuring at least three points, and pressing Terminate
	Axis B	Measuring at least three points, and pressing Terminate
	Hole C	Measuring at least three points, and pressing Terminate
		X-Z flat surface specification (2nd axis Z axis)
		Rotation projection mode "ON"
		Probe calibration
		Measuring at least any three points on the R of the D part, and pressing Terminate.
		Rotation projection mode "OFF"
		Automatic detection

```
A0001 A05 PLN Z -66.589
AVXY -52.08.44 AVZ 0.00.28
```

```
C0001 A80 COD2 Reference surface set (XY)
```

```
C0002 A81 COD2 Origin point set
```

```
A0002 A02 CIR Diameter 60.045 Radius 30.022
X -122.864 Y -159.719
```

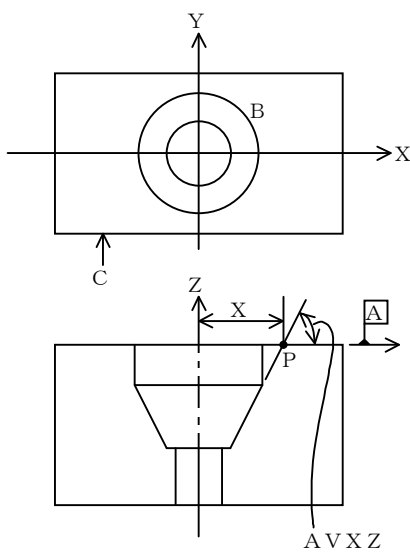
```
C0003 A81 COD2 Origin point set
```

```
A0003 A02 CIR Diameter 12.010 Radius 6.005
X 13.045 Y -0.107
```

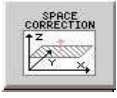
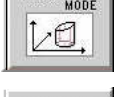
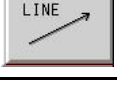
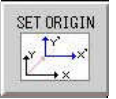
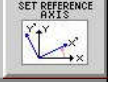
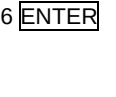
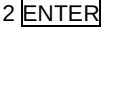
```
C0004 A82 COD2 Reference axis set (X)
```

```
A0004 A02 CIR Diameter 159.894 Radius 79.947
X 107.433 Z -30.002
```

2) Taper measurement (when using the rotation projection mode)



* Crossing angle (AVXZ) is an angle formed with the reference axis, and X is a taper radius on the reference surface.

Process code	Position	Measurement method
      	Surface A	Measuring at least three points, and pressing Terminate Measuring at least three points, and pressing Terminate
  	Hole B	Measuring at least three points, and pressing Terminate
 	Edge C	Measuring at least two points, and pressing Terminate
		X-Z flat surface specification (measurement axis Z axis)
		Rotation projection specification
		Measuring at least any two points on the taper, and pressing Terminate.

C0001 A80 COD2 Reference surface set(X Y)

C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 28.060
X -343.097 Y -123.034

C0003 A81 COD2 Origin point set

A0003 A01 LIN Y -50.050
AVXY 2.559

C0004 A82 COD2 Reference axis set(X)

A0004 A01 LIN X 14.963
AVXZ 82.443

22. How to use PH-9

1) Calibration

Examine the direction of probe suitable for the workpiece in advance.

- (1)  Probe No. (*,*, ...*.*): [0]: Input "0.2", and press **ENTER**.

(2) Use the HCU1 to make the probe direction downward.

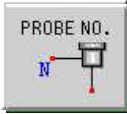
(3) As it is in the calibration mode, measure at least four points in the starting point block, and press Terminate.

Now, the registration of probe No.0 is finished.

As the calibration can be executed continuously, repeat the steps ② and ③ for the direction of the other probes. Up to 100 probes can be registered.

2) Operation for manual measurement and teaching measurement

After a calibration, it is possible to execute measurement by specifying a probe No. you want to use even while the manual measurement or the teaching measurement is underway.

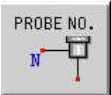
If the probe No. is specified by , the probe direction of PH-9 can be changed. As it is displayed as "Press **ENTER** key, move the probe to where it does not interfere anything for changing the direction, and press **ENTER**."

MEMO

23. Auto calibration

- 1) After the system startup, execute the calibration for the No.0 probe.
- 2) Execute the calibration teaching measurement.

Example) For executing a teaching measurement for probes of two directions using PH-9

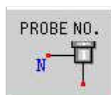
- (1)  Part program name: ____
- (2)  Probe number: Input "0", and press , message display, press .
- (3)  Probe number: Input "0", and press .

Set downward for the probe No.0 (AO_BO).

The teaching measurement is executed while inputting middle points by using the No.0 probe.

Note: Move the probe to where it does not interfere anything even the next probe (No.1) rotates, and press Terminate.

Next, in order to register the No.1 probe, change the probe No. to "1."

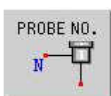
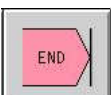
- (4)  Probe number: Input "1", and press , message display, press .

If the probe direction is different, change the PH-9 angle to rotate the probe, and input the middle point at that position.

Or, rotate the probe by using HCU1, and input the middle point at that position (The angle and the middle point of PH-9 are registered).

- (5)  Probe number: Input "1", and press
The teaching measurement is executed while inputting middle points by using the No.1 probe.

Note: Move the probe to where it does not interfere anything even rotating the No.0 probe, and press Terminate for the last probe.

- (6)  Probe number: Input "0", and press , message display, press .
- (7)  Exit of teaching

3) Auto calibration is executed.



Part program name: _____
Execution stop line

- * The middle point, probing point, and terminate coordinate value during the calibration are registered in the 1st coordinate system.
- * Do not use the probe information initialization. If the probe information initialization is executed mistakenly before executing the auto calibration, never fail to execute the manual calibration or the semi-auto calibration for all the probes.

24. Example of shape deviation, posture deviation, and positional deviation

Shape deviation

(1) Flatness ()

Measure at least four points, and press Terminate.

A05	PLN	Flatness	0.003	Maximum	0.001	Minimum value	-0.001
		Standard deviation	0.001	value			

(2) Roundness ()

Measure at least four points, and press Terminate.

A02	CIR	Diameter	28.012	Radius	14.006	Roundness	0.033
		Maximum Value	14.023	Minimum value	13.989	Standard deviation	0.011

(3) Straightness ()

Measure at least three points, and press Terminate.

A01	LIN	Straightness	0.002	Maximum value	0.001	Minimum value	-0.001
		Standard deviation	0.001				

(4) Cylindricity ()

Measure at least seven points, and press Terminate.

A06	CYL	Diameter	11.995	Radius	5.998	Cylindricity	0.017
		Maximum Value	6.005	Minimum value	5.988	Standard deviation	0.005

(5) Sphericity ()

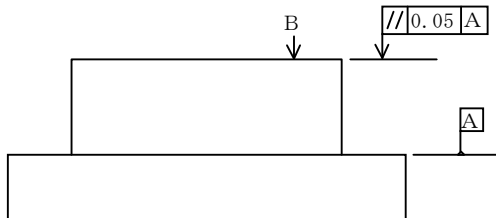
Measure at least five points, and press Terminate.

A04	SPR	Diameter	25.398	Radius	12.699	Sphericity	0.016
		Maximum Value	12.711	Minimum value	12.694	Standard deviation	0.005

Posture deviation

(1) Parallelism (//)

A. Surface against reference surface



	A  
	B Measure at least two points, and press Terminate.

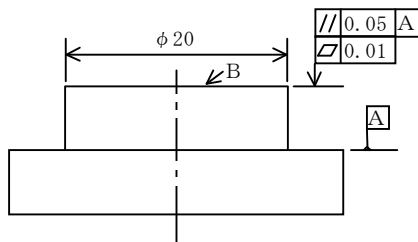
A0001 A05 PLN Flatness 0.003

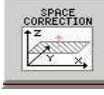
C0001 A80 COD2 Reference surface set(XY)

C0002 A81 COD2 Origin point set

A0002 A00 PNT Z 8.117
Parallelism Z 0.031

B. Surface against reference surface (flatness + parallelism)



	A  
	B Measure at least four points, and press Terminate.
	Datum: Input "2", and press ENTER (coordinate system flat surface). MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Size of surface: Input "20", and press ENTER

A0001 A05 PLN Flatness 0.013

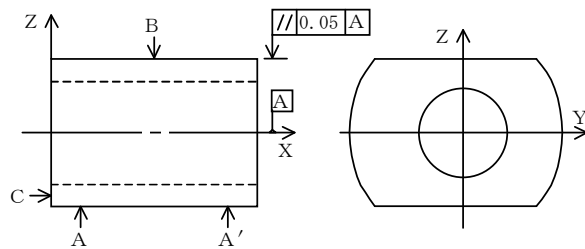
C0001 A80 COD2 Reference surface set(XY)

C0002 A81 COD2 Origin point set

A0002 A05 PLN Flatness 0.003

A54 Parallelism Z 0.028

C. Surface against reference axis



A	
B	
C	
	B Measure at least two points before and after, and press Terminate.

```
A0001 A06 CYL Diameter      27.970
          AVYZ      -160.40.57  AVX      0.13.48
```

```
C0001 A80 COD2 Reference surface set (YZ)
```

```
C0002 A81 COD2 Origin point set
```

```
A0002 A01 LIN Z              29.959
```

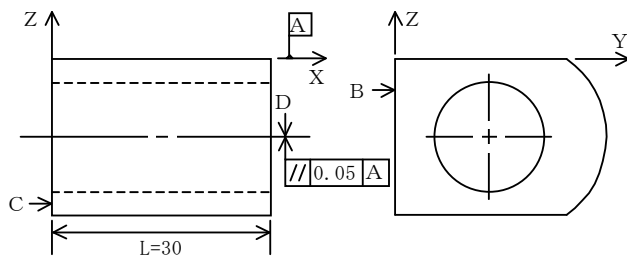
```
C0003 A82 COD2 Reference axis set (Y)
```

```
A0003 A00 PNT X              -0.032
```

```
C0004 A81 COD2 Origin point set
```

```
A0004 A00 PNT Z              29.931
          Parallelism Z      0.004
```

D. Axis against reference surface



A	
B	
C	
D	
	Datum: Input "2", and press ENTER (coordinate system flat surface). Coordinate system flat surface: Input "1", press ENT (XY). MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis length: Input "30", and press ENTER.

```

A0001 A05 PLN Flatness 0.002

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A01 LIN Y -38.025

C0003 A82 COD2 Reference axis set (X)

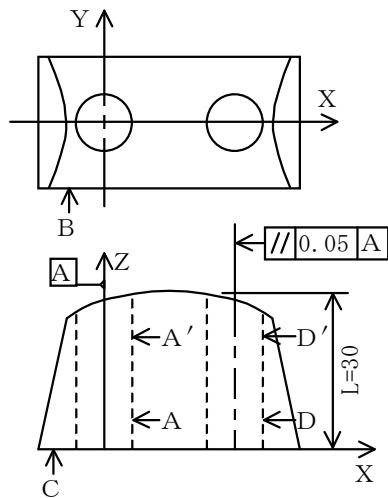
A0003 A00 PNT X -58.173

C0004 A81 COD2 Origin point set

A0004 A06 CYL Diameter 27.949
AVYZ 131.01.21 AVX 0.04.12

A54 Parallelism Z 0.028
  
```

E. Axis against reference axis



CYLINDER	AA'	SPACE CORRECTION	SET ORIGIN
STRAIGHT LINE	B	SET REFERENCE AXIS	
POINT	C	SET ORIGIN	
CYLINDER	DD'		
PARALLELISM	Datum: Input "1", and press ENTER (coordinate axis). MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis length: Input "30", and press ENTER . Tolerance region: Input "0", and press ENTER (circle).		

```

A0001 A06 CYL Diameter      11.988
      AVXY      -121.13.53   AVZ   0.04.52

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A01 LIN  Y             -37.578
      AVXY      -0.33.19

C0003 A82 COD2 Reference axis set (X)

A0003 A00 PNT  Z             37.550

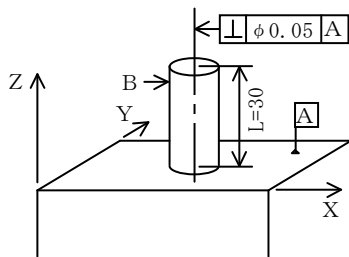
C0004 A81 COD2 Origin point set

A0004 A06 CYL Diameter      11.993
      AVYZ      53.07.24    AVZ   0.04.58

A54      Parallelism      0.014
  
```

(2) Squareness () $\perp$

A. Axis against reference surface



A	
B	
	Datum: Input "2", and press ENTER (coordinate system flat surface). MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis length: Input "30", and press ENTER. Tolerance region: Input "0", and press ENTER (circle).

A0001 A05 PLN Flatness 0.000

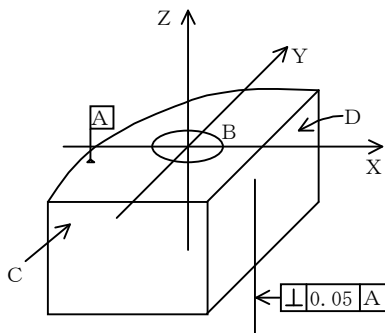
C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A06 CYL Diameter 11.986
AVXY 155.50.54 AVZ 0.14.37

A55 Squareness 0.043

A. Surface against reference surface



A	
PLANE	SPACE CORRECTION SET ORIGIN
B	
CIRCLE	SET ORIGIN
C	
STRAIGHT LINE	SET REFERENCE AXIS
POINT	D Measure at least two points, and press Terminate.

```

A0001 A05 PLN Flatness 0.000

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 28.019
X 11.172 Y -27.283

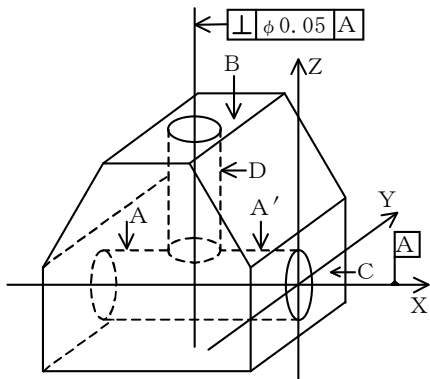
C0003 A81 COD2 Origin point set

A0003 A01 LIN Straightness 0.001

C0004 A82 COD2 Reference axis set (X)

A0004 A00 PNT X 30.181
Parallelism X 0.011 (the same meaning as
squareness)
  
```

C. Axis against reference axis



AA'	
B	
C	
D	
	Datum: Input "1", and press ENTER (coordinate axis). Coordinate axis: Input "1", and press ENTER (X axis) MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis length: Input "30", and press ENTER .

```

A0001 A06 CYL Diameter          27.940
      AVYZ                      179.52.18   AVX 0.34.56

C0001 A80 COD2 Reference surface set (YZ)

C0002 A81 COD2 Origin point set

A0002 A01 LIN Straightness       0.004

C0003 A82 COD2 Reference axis set (Y)

A0003 A00 PNT X                  41.214

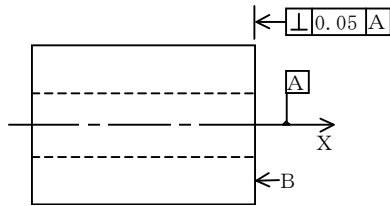
C0004 A81 COD2 Origin point set

A0004 A06 CYL Diameter          11.994
      AVXY                      -175.10.50   AVZ 0.08.51

A55      Squareness X 0.026

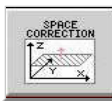
```

D. Surface against reference axis





CYLINDER



SPACE CORRECTION



SET ORIGIN



POINT

B Measure at least two points, and press Terminate (parallelism X= squareness).

Parallelism against Y-Z surface

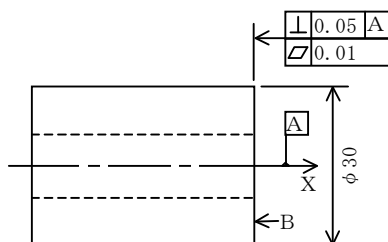
```
A0001 A06 CYL Diameter      27.933
      AVYZ      179.12.34   AVX    0.34.08
```

```
C0001 A80 COD2 Reference surface set (YZ)
```

```
C0002 A81 COD2 Origin point set
```

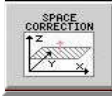
```
A0002 A00 PNT X              41.207
      Parallelism X  0.036
```

E. Surface against reference axis (flatness + squareness)





CYLINDER



SPACE CORRECTION

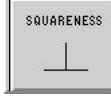


SET ORIGIN



PLANE

B



SQUARENESS

Datum: Input "1", and press **ENTER** (coordinate axis).
MMC/LMC mode: Input "0" and "0", and press **ENTER** (no application).
Size of surface: Input "30", and press **ENTER**

```
A0001 A06 CYL Diameter      27.934
      AVYZ      178.34.30   AVX    0.32.51
```

```
C0001 A80 COD2 Reference surface set (YZ)
```

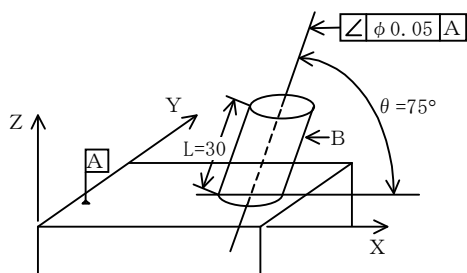
```
C0002 A81 COD2 Origin point set
```

```
A0002 A05 PLN Flatness       0.003
```

```
A55      Squareness         0.031
```


(3) Gradient (∠)

A. Axis against reference surface



PLANE

A

SPACE CORRECTION

SET ORIGIN

CYLINDER

B

ANGULARITY

Datum: Input "2", and press (coordinate system flat surface).
Coordinate system flat surface: Input "1", press (XY).
MMC/LMC mode: Input "0" and "0", and press ENTER (no application).
Axis length: Input "30", and press .
Angle of slope : 75

```
A0001 A05 PLN Flatness      0.000

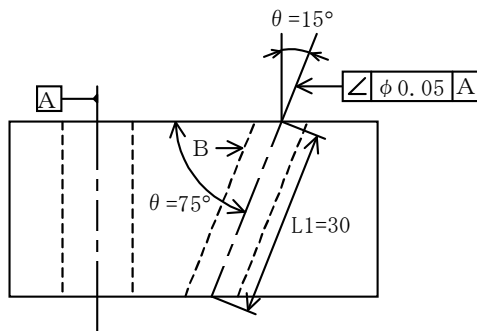
C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A06 CYL Diameter      14.037
      AVXY      -0.14.10  AVZ      15.11.42

      A56      Gradient      0.068
```

B. Axis against reference axis



A	
B	
	Datum: Input "1", and press ENTER (coordinate axis). Coordinate axis: Input "1", and press ENTER (X axis) MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis length: Input "30", and press ENTER . Angle of slope : 15 ENTER

```

A0001 A06 CYL Diameter    11.989
          AVXY      -21.59.00  AVZ          0.02.16

C0001 A80 COD2 Reference surface set (XY)

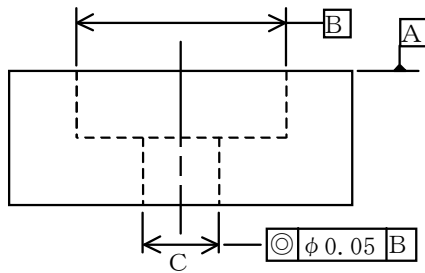
C0002 A81 COD2 Origin point set

A0002 A06 CYL Diameter    14.027
          AVXY      -0.15.16  AVZ          15.10.23

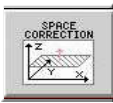
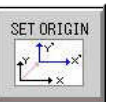
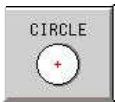
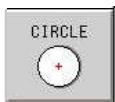
          A56      Gradient    0.060
  
```

Positional deviation

(1) Concentricity (◎)



Method 1

	A		
	B		
	C		
	MMC/LMC mode: Input "0" and "0", and press ENTER (no application).		

```

A0001 A05 PLN Flatness 0.001

C0001 A80 COD2 Reference surface set (XY)

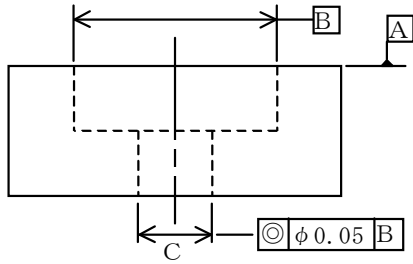
C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 28.012
      X          9.073 Y -27.588
      Roundness 0.022

C0003 A81 COD2 Origin point set

A0003 A02 CIR Diameter 27.212
      X          0.009 Y 0.001
      Roundness 0.016

      A52 Concentricity 0.018
  
```



Method 2

	A		
	B		
	C		Concentricity : radius vector R×2

```

A0001 A05 PLN    Flatness           0.007

C0001 A80 COD2   Reference surface set (XY)

C0002 A81 COD2   Origin point set

A0002 A02 CIR    Diameter           22.032
                  X                  -10.719      Y      -44.939
                  Roundness          0.003

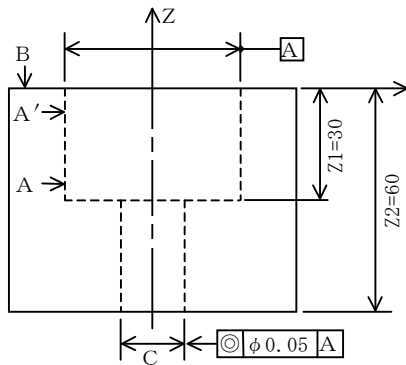
C0003 A81 COD2   Origin point set

A0003 A02 CIR    Diameter           13.007
                  X                  -0.001      Y        0.032
                  Roundness          0.002

A0003 B38 CIR    Radius vector R 0.032      AXY 91.10.39  Z    0.000
  
```

(2) Coaxiality (◎)

A. Axis against reference axis



AA'	
B	
C	
MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis edge (Z1, Z2): Input "-30", and "-60", and press ENTER.	

```

A0001 A06 CYL Diameter    22.029
          AVXY      35.52.21  AVZ      0.01.49

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A00 PNT Z          220.091

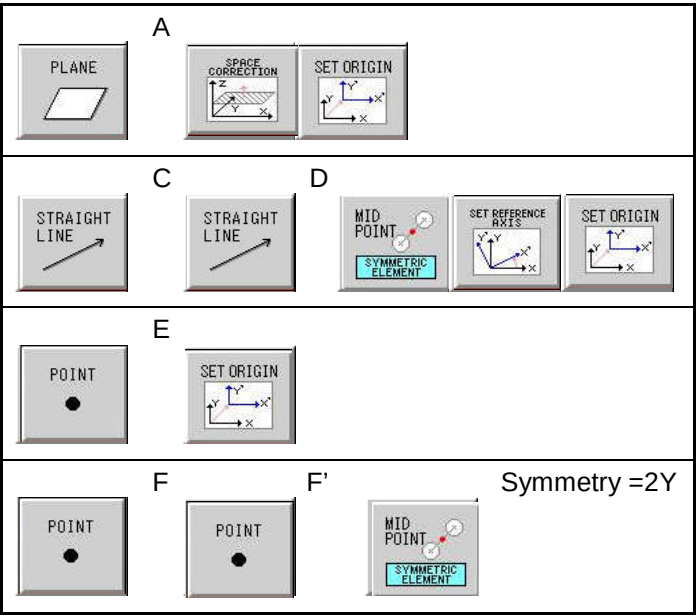
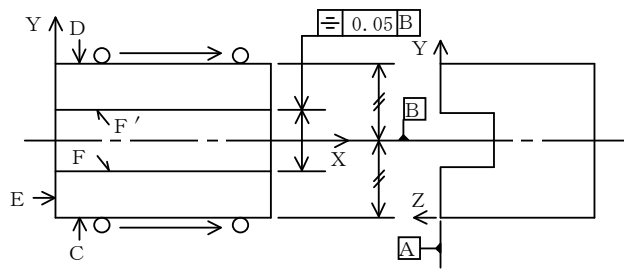
C0003 A81 COD2 Origin point set

A0003 A06 CYL Diameter    13.009
          AVXY     -156.11.09  AVZ      0.05.05

A52      Coaxiality 0.111
  
```

(3) Symmetry (≡)

A.



```
A0001 A05 PLN Flatness 0.005

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A01 LIN Straightness 0.005

A0003 A01 LIN Straightness 0.000

A0004 A62 LIN Y -29.920 Z 0.000
AVYZ 0.00.00 AVX 4.02.59

C0003 A82 COD2 Reference axis set (X)

C0004 A81 COD2 Origin point set

A0005 A00 PNT X 2.101

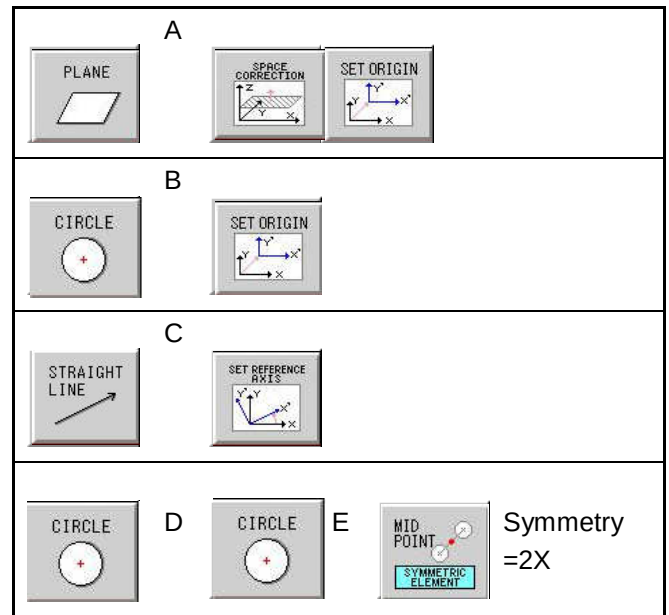
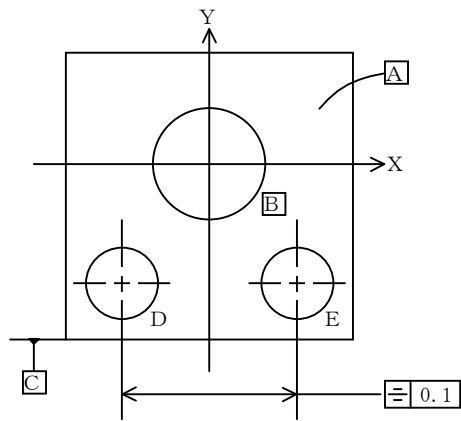
C0005 A81 COD2 Origin point set

A0006 A00 PNT Y -12.587

A0007 A00 PNT Y 12.339

A0008 A62 PNT Distance 24.925
X 0.000 Y -0.124 Z 0.000
```

B.



A0001 A05 PLN Flatness 0.001

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 28.013
X 11.102 Y -28.756
Roundness 0.016

C0003 A81 COD2 Origin point set

A0003 A01 LIN Y -50.085
AVXY 1.24.20
Straightness 0.001

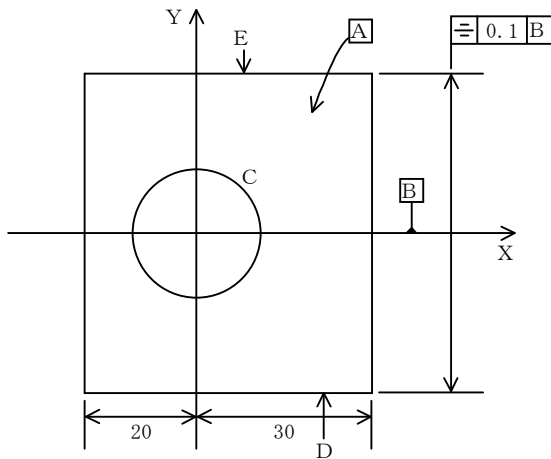
C0004 A82 COD2 Reference axis set (X)

A0004 A02 CIR Diameter 11.996
X -21.641 Y -12.490

A0005 A02 CIR Diameter 11.998
X 21.658 Y -12.499

A0006 A62 PNT Distance 43.299
X 0.009 Y -12.494 Z 0.000

C.



	A
	C
	D
	E
	MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Axis edge (X1, X2): Input "-20", and "30", and press ENTER.

A0001 A05 PLN Flatness 0.000

C0001 A80 COD2 Reference surface set (XY)

C0002 A81 COD2 Origin point set

A0002 A02 CIR Diameter 10.542
X 39.613 Y -26.912
Roundness 0.041

C0003 A81 COD2 Origin point set

A0003 A01 LIN Y -12.564
AVXY 4.35.47
Straightness 0.007

C0004 A82 COD2 Reference axis set (X)

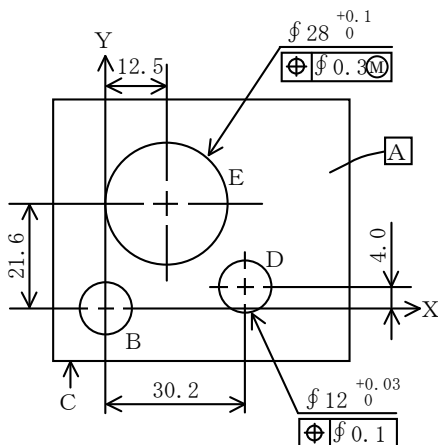
A0004 A01 LIN Y 12.377
AVXY -0.04.40
Straightness 0.009

A0005 A62 LIN Y -0.073 Z 0.000
AVYZ 180.00.00 AVX 0.02.20

A53 Symmetry Y 0.174 SymmetryZ 0.000

(4) Degree of location ($\oplus$) MMC mode application: Yes, No

① For $\oplus \phi 0.1$



	A	
	B	
	C	
	D	
	Datum address: Input "0", and press ENTER (coordinate system starting point). Nominal value X: Input "30" and "2", and press ENTER. Nominal value Y: Input "4", and press ENTER. Nominal value Z: Input "0", and press ENTER. MMC/LMC mode: Input "0" and "0", and press ENTER (no application). Tolerance region: Input "0", and press ENTER (circle).	

② For $\oplus \phi 0.3M$

	1 ENT (nominal value collation: ON)
	E (for the diameter, check the nominal value) Diameter key code: 3 ENTER Label : CIR ENTER Nominal value : 28.0000 ENTER Upper limit tolerance: 0.1000 ENTER Lower limit tolerance: 0.0000 ENTER Radius key code : 0 ENTER X key code : 1 ENTER Label : ENTER Y key code : 1 ENTER Label : ENTER

(continue)



Datum address: Input "0", and press

Nominal value X: Input "12.5", and press

Nominal value Y: Input "21.6", and press

Nominal value Z: Input "0", and press

MMC/LMC mode: Input "1", and "0", and press ENTER
(applying MMC for the element immediately before).

Tolerance region: Input "0", and press ENTER (circle).

Location degree - Key code: Input "3", and press ENTER.

Label:

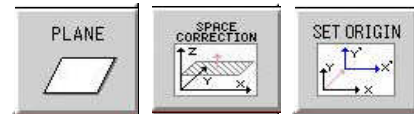
Upper limit tolerance: 0.3000

A0001	A05	PLN	Flatness	0.001				
C0001	A80	COD2	Reference surface set(XY)					
C0002	A81	COD2	Origin point set					
A0002	A02	CIR	Diameter	12.001				
			X	-3.710	Y		-51.623	
			Roundness	0.011				
C0003	A81	COD2	Origin point set					
A0003	A01	LIN	Y	-8.511				
			AVXY	0.53.27				
			Straightness	0.001				
C0004	A82	COD2	Reference axis set(X)					
A0004	A02	CIR	Diameter	11.995				
			X	30.174	Y		3.965	
			Roundness	0.014				
MMC No	A51		<u>Degree of location</u>	<u>0.087</u>				
A0005	A02	CIR	Diameter	28.015	28.000	0.015	0.100	-0.000 +
			X	12.500	Y		21.659	
			Roundness	0.023				
MMC Yes	A51		<u>Degree of location</u>	<u>M</u>		0.118	0.315(0.300)	++

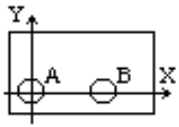
MEMO

25. Example of setting coordinate system

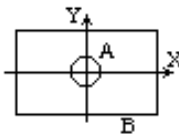
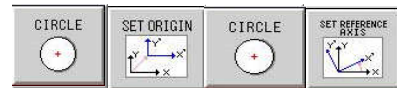
1. If the primary reference is flat surface, inclination of the reference surface is corrected.



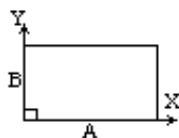
For the subsequent measurement methods, refer to the description below.



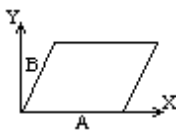
Assist 1



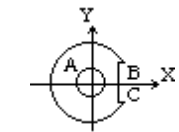
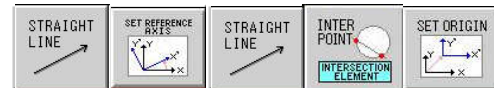
Assist 2



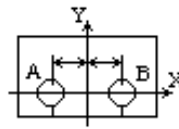
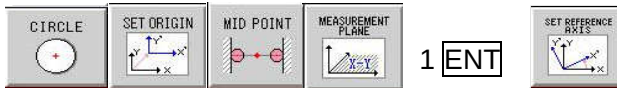
Assist 3



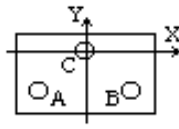
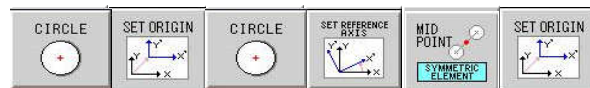
Assist 4



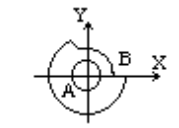
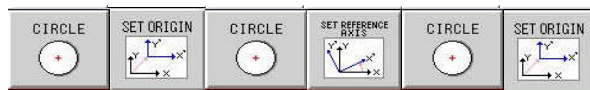
Assist 5



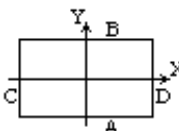
Assist 6



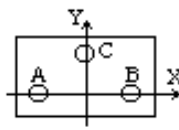
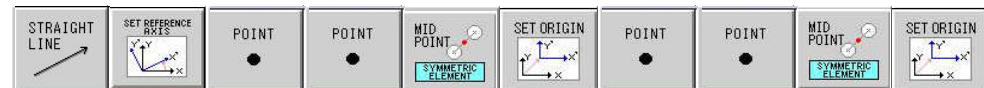
Assist 7



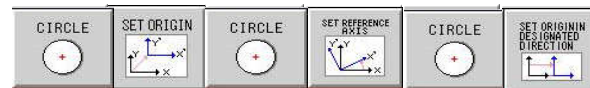
Assist 8

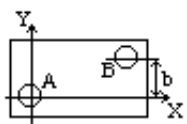


Assist 9

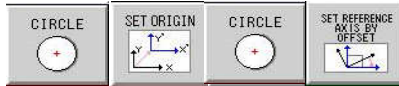


Assist 10

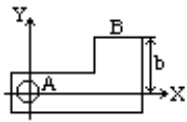




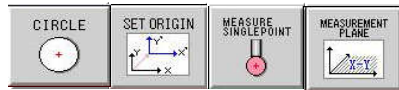
Assist 11



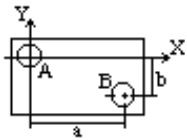
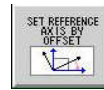
2 b



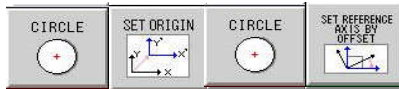
Assist 12



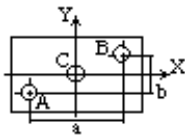
1 2 b + probe radius



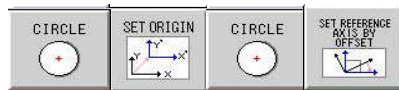
Assist 13



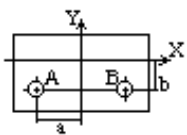
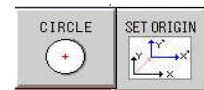
4 a -b



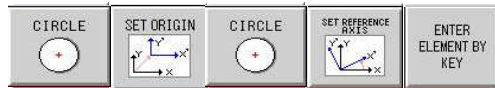
Assist 14



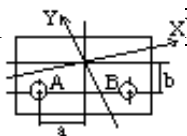
4 a b



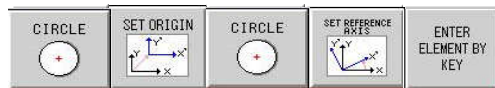
Assist 15



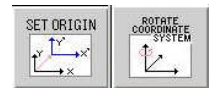
1 0 a,b,0



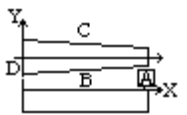
Assist 16



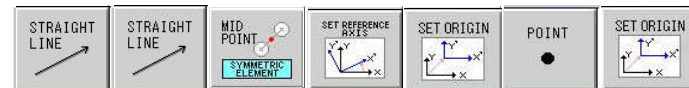
1 0 a,b,0



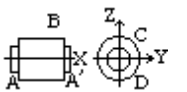
1 0



Assist 17



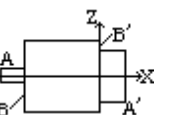
2. If the primary reference is axis center



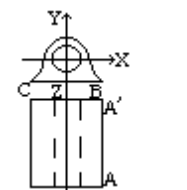
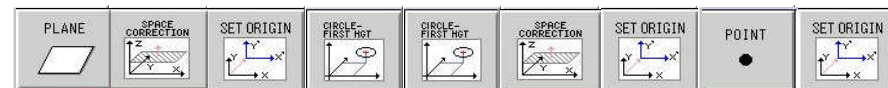
Assist 18



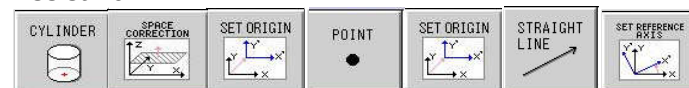
2



Assist 19



Assist 20



- Inquiry concerning measuring method -

〈 Sent to 〉 TOKYO SEIMITSU CO.,LTD

	TEL No.	FAX No.
☐ Tsuchiura factory application team	029-831-1234	029-831-6623
☐ Nagoya show room	0561-32-8501	0561-32-8618
☐ Osaka show room	06-6821-0221	06-6821-0210

〈 Question 〉

Content	Schematic drawing
Attached drawing: Yes, No	

Company Name: _____

Department: _____ Name: _____

TEL : _____ FAX : _____

Model of the measurement machine: _____

Software:XYANA2000 Virtual PC _____

Probe: _____

Delivery day: _____